

INTEGRAL TRANSFORMS

20.1 INTRODUCTION

Frequently in mathematical physics we encounter pairs of functions related by an expression of the form

$$g(x) = \int_a^b f(t)K(x, t)dt, \quad (20.1)$$

where it is understood that a , b , and $K(x, t)$ (called the **kernel**) will be the same for all function pairs f and g . We can write the relationship expressed in Eq. (20.1) in the more symbolic form

$$g(x) = \mathcal{L}f(t), \quad (20.2)$$

thereby emphasizing the fact that Eq. (20.1) can be interpreted as an operator equation. The function $g(x)$ is called the integral transform of $f(t)$ by the operator $\mathcal{L}$, with the specific transform determined by the choice of a , b , and $K(x, t)$. The operator defined by Eq. (20.1) will be linear:

$$\int_a^b [f_1(t) + f_2(t)]K(x, t)dt = \int_a^b f_1(t)K(x, t)dt + \int_a^b f_2(t)K(x, t)dt, \quad (20.3)$$

$$\int_a^b cf(t)K(\alpha, t)dt = c \int_a^b f(t)K(\alpha, t)dt. \quad (20.4)$$

In order for transforms to be useful, we will shortly see that we need to be able to “undo” their effect. From a practical viewpoint, this means that not only must there exist

an operator $\mathcal{L}^{-1}$, but also that we have a reasonably convenient and powerful method of evaluating

$$\mathcal{L}^{-1}g(x) = f(t) \quad (20.5)$$

for an acceptably broad range of $g(x)$. The procedure for inverting a transform takes a wide variety of forms that depend on the specific properties of $K(x, t)$, so we cannot write a formula that is as general as that for $\mathcal{L}$ in Eq. (20.1).

Not all superficially reasonable choices for the kernel $K(x, t)$ will lead to operators $\mathcal{L}$ that have inverses, and even for strategically chosen kernels it may be the case that $\mathcal{L}$ and $\mathcal{L}^{-1}$ will only exist for substantially restricted classes of functions. Thus, the entire development of the present chapter is restricted (for any given integral transform) to functions for which the indicated operations can be carried out.

Before embarking on a study of integral transforms, we may well ask, “Why are integral transforms useful?” Their most common applications are in situations illustrated schematically in Fig. 20.1, where we have a problem that can be solved only with difficulty, if at all, in its original formulation (usually in ordinary space, sometimes called **direct** or **physical space**). However, it may happen that the transform of the problem can be solved relatively easily. Our strategy, then, will be to formulate and solve our problem in the transform space, after which we transform the solution back to direct space. This strategy often works because the most popular integral transforms are changed in simple ways by differentiation and integration operators, with the result that differential and integral equations assume relatively simple forms. This feature will be discussed and illustrated at length later in this chapter.

Another frequent use of integral transforms is to use one, together with its inverse, to form an **integral representation** of a function that we originally had in an explicit form. This move (which appears to be in the direction of generating greater complexity) has value that arises from the relatively simple behavior of the transforms of differentiation and integration operators. Procedures involving integral representations are also presented in later sections of this chapter.

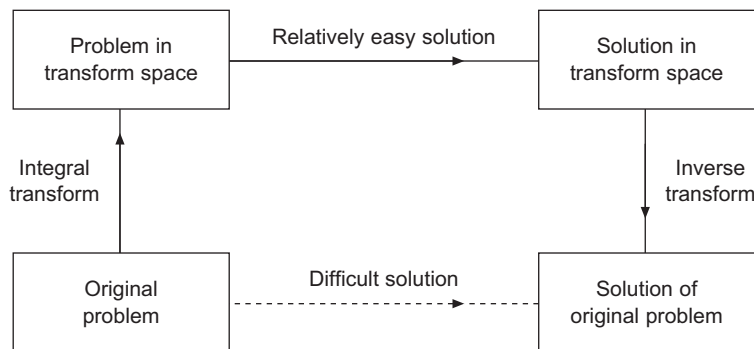


FIGURE 20.1 Schematic: use of integral transforms.

Some Important Transforms

The integral transform that has seen the widest use is the **Fourier transform**, defined as

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt. \quad (20.6)$$

The notation for this transform is not entirely universal; some writers omit the prefactor $1/\sqrt{2\pi}$; we keep it because it causes the transform and its inverse to have formulas that are more symmetrical. In applications involving periodic systems, one occasionally encounters a definition with kernel $\exp(2\pi i\omega t/a_0)$, where a_0 is a lattice constant. These differences in notation do not change the mathematics, but cause formulas to differ by powers of 2π or a_0 . Caution is therefore advised when combining material from different sources.

We have defined the Fourier transform in a notation that assigns the symbol ω to the transform variable. We did so because, in studying signal processing (an important use of Fourier transforms), the function $f(t)$ usually represents the time behavior of a signal (typically a wave distribution of some kind). Its Fourier transform, $g(\omega)$, can then be identified as the corresponding frequency distribution. However, it is worth pointing out that Fourier transforms turn up in contexts far removed from signal-processing problems; they can be used to advantage in evaluating integrals, in alternative formulations of quantum mechanics, and in a wide range of other mathematical procedures.

A second transform that has historically been of great importance is the **Laplace transform**,

$$F(s) = \int_0^{\infty} e^{-ts} f(t) dt. \quad (20.7)$$

One of its useful features is the fact that under transformation, differential equations become algebraic equations (as we shall see in detail in [Section 20.8](#)). Since algebraic equations are usually easier to solve than differential equations, this feature lends itself to the strategy illustrated in [Fig. 20.1](#). A disadvantage of the Laplace transform is that the formula for its inverse is relatively difficult to use. Historically, this difficulty was dealt with by developing tables of Laplace transforms (which can be used to identify inverses). As digital computers have become more powerful, the use of Laplace transforms has declined, but they remain sufficiently useful that we treat them in some detail in the present chapter.

Among other transforms that have seen significant use, we mention here two:

1. The **Hankel transform**,

$$g(\alpha) = \int_0^{\infty} f(t) t J_n(\alpha t) dt. \quad (20.8)$$

This transform represents the continuum limit of the Bessel series we studied in Eqs. (14.47) and (14.48).

2. The **Mellin transform**,

$$g(\alpha) = \int_0^{\infty} f(t) t^{\alpha-1} dt. \quad (20.9)$$

We have actually used the Mellin transform without calling it by name; for example, $g(\alpha) = \Gamma(\alpha)$ is the Mellin transform of $f(t) = e^{-t}$. Many Mellin transforms are given in a text by Titchmarsh (see Additional Readings).

20.2 FOURIER TRANSFORM

We proceed now to a more detailed discussion of the Fourier transform,

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt. \quad (20.10)$$

If we rewrite the exponential in Eq. (20.10) in terms of the sine and cosine, and then restrict consideration to functions that are assumed to be either even or odd functions of x , we obtain variants of the original form that are also useful integral transforms:

$$g_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos \omega t dt, \quad (20.11)$$

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin \omega t dt. \quad (20.12)$$

These formulas define the **Fourier cosine** and **Fourier sine** transforms. Their kernels, which are real, are natural for use in studies of wave motion and for extracting information from waves, particularly when phase information is involved. The output of a stellar interferometer, for instance, involves a Fourier transform of the brightness across a stellar disk. The electron distribution in an atom may be obtained from a Fourier transform of the amplitude of scattered x-rays.

Example 20.2.1 SOME FOURIER TRANSFORMS

1. $f(t) = e^{-\alpha|t|}$, with $\alpha > 0$. To deal with the absolute value, we break the transform integral into two regions:

$$\begin{aligned} g(\omega) &= \sqrt{\frac{1}{2\pi}} \int_{-\infty}^0 e^{\alpha t + i\omega t} dt + \sqrt{\frac{1}{2\pi}} \int_0^{\infty} e^{-\alpha t + i\omega t} dt \\ &= \sqrt{\frac{1}{2\pi}} \left[\frac{1}{\alpha + i\omega} + \frac{1}{\alpha - i\omega} \right] = \sqrt{\frac{1}{2\pi}} \frac{2\alpha}{\alpha^2 + \omega^2}. \end{aligned} \quad (20.13)$$

We note two features of this result: (1) It is real; from the form of the transform, we can see that if $f(t)$ is even, its transform will be real. (2) The more localized is $f(t)$, the less localized will be $g(\omega)$. The transform will have an appreciable value until $\omega \gg \alpha$; larger α corresponds to greater localization of $f(t)$.

2. $f(t) = \delta(t)$. We easily find

$$g(\omega) = \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} \delta(t) e^{i\omega t} dt = \sqrt{\frac{1}{2\pi}}. \quad (20.14)$$

This is the ultimately localized $f(t)$, and we see that $g(\omega)$ is completely delocalized; it has the same value for all ω .

3. $f(t) = 2\alpha\sqrt{1/2\pi}/(\alpha^2 + t^2)$, with $\alpha > 0$. One way to evaluate this transform is by contour integration. It is convenient to start by writing initially

$$g(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\alpha e^{i\omega t}}{(t - i\alpha)(t + i\alpha)} dt.$$

The integrand has two poles: one at $t = i\alpha$ with residue $e^{-\alpha\omega}/i$ and one at $t = -i\alpha$ with residue $e^{+\alpha\omega}/(-i)$. If $\omega > 0$, our integrand will become negligible on a large semicircle in the upper half-plane, so an integral over the contour shown in Fig. 20.2(a) will be that needed for $g(\omega)$. This contour encloses only the pole at $t = i\alpha$, so we get

$$g(\omega) = \frac{1}{2\pi} (2\pi i) \frac{e^{-\alpha\omega}}{i} \quad (\omega > 0). \quad (20.15)$$

However, if $\omega < 0$, we must close the contour in the lower half-plane, as in Fig. 20.2(b), circling the pole at $t = -i\alpha$ in a clockwise sense (thereby generating a minus sign). This procedure yields

$$g(\omega) = \frac{1}{2\pi} (-2\pi i) \frac{e^{+\alpha\omega}}{-i} \quad (\omega < 0). \quad (20.16)$$

If $\omega = 0$, we cannot perform a contour integration on either of the paths shown in Fig. 20.2, but we then do not need this sophisticated an approach, as we have the

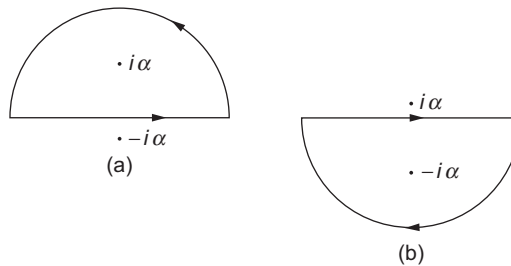


FIGURE 20.2 Contours for third transform in Example 20.2.1.

elementary integral

$$g(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\alpha}{t^2 + \alpha^2} dt = 1. \quad (20.17)$$

Combining Eqs. (20.15)–(20.17) and simplifying, we have

$$g(\omega) = e^{-\alpha|\omega|}.$$

Here we Fourier transformed the transform from our first example, recovering the original untransformed function. This provides an interesting clue as to the form to be expected for the inverse Fourier transform. It is only a clue, because our example involved a transform that was real (i.e., not complex). ■

An important Fourier transform follows.

Example 20.2.2 FOURIER TRANSFORM OF GAUSSIAN

The Fourier transform of a Gaussian function e^{-at^2} , with $a > 0$,

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-at^2} e^{i\omega t} dt,$$

can be evaluated analytically by completing the square in the exponent,

$$-at^2 + i\omega t = -a \left(t - \frac{i\omega}{2a} \right)^2 - \frac{\omega^2}{4a},$$

which we can check by evaluating the square. Substituting this identity and changing the integration variable from t to $s = t - i\omega/2a$, we obtain (in the limit of large T)

$$g(\omega) = \frac{1}{\sqrt{2\pi}} e^{-\omega^2/4a} \int_{-T-i\omega/2a}^{T-i\omega/2a} e^{-as^2} ds. \quad (20.18)$$

The s integration, shown in Fig. 20.3, is on a path parallel to, but below the real axis by an amount $i\omega/2a$. But because connections from that path to the real axis at $\pm T$ make negligible contributions to a contour integral and since the contours in Fig. 20.3 enclose no

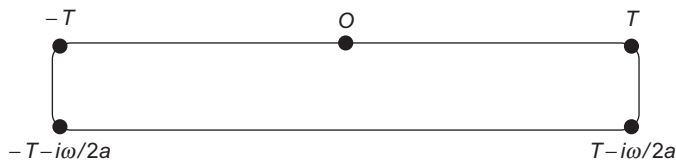


FIGURE 20.3 Contour for transform of Gaussian in Example 20.2.2.

singularities, the integral in Eq. (20.18) is equivalent to one along the real axis. Changing the integration limits to $\pm\infty$ and rescaling to the new variable $\xi = s/\sqrt{a}$, we reach

$$\int_{-\infty}^{\infty} e^{-as^2} dt = \frac{1}{\sqrt{a}} \int_{-\infty}^{\infty} e^{-\xi^2} d\xi = \sqrt{\frac{\pi}{a}},$$

where we have used Eq. (1.148) to evaluate the error-function integral. Substituting these results we find

$$g(\omega) = \frac{1}{\sqrt{2a}} \exp\left(-\frac{\omega^2}{4a}\right), \quad (20.19)$$

again a Gaussian, but in ω -space. An increase in a makes the original Gaussian e^{-at^2} narrower, while making wider its Fourier transform, the behavior of which is dominated by the exponential $e^{-\omega^2/4a}$. ■

Fourier Integral

When we first encountered the delta function, its representation which is the large- n limit of

$$\delta_n(t) = \frac{1}{2\pi} \int_{-n}^n e^{i\omega t} d\omega, \quad (20.20)$$

was identified as particularly useful in Fourier analysis. We now use that representation to obtain an important result known as the **Fourier integral**. We write the fairly obvious equation,

$$\begin{aligned} f(x) &= \lim_{n \rightarrow \infty} \int_{-\infty}^{\infty} f(t) \delta_n(t-x) dt \\ &= \lim_{n \rightarrow \infty} \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t) \left[\int_{-n}^n e^{i\omega(t-x)} d\omega \right] dt. \end{aligned} \quad (20.21)$$

We now interchange the order of integration and take the limit $n \rightarrow \infty$, reaching

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} d\omega \int_{-\infty}^{\infty} dt f(t) e^{i\omega(t-x)}.$$

Finally, we rearrange this equation to the form

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega x} d\omega \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt. \quad (20.22)$$

Equation (20.22), the **Fourier integral**, is an integral representation of $f(x)$, and will be more obviously recognized as such if the inner integration (over t) is performed, leaving

unevaluated that over ω . In fact, if we identify the inner integration as (apart from a factor $\sqrt{1/2\pi}$) the Fourier transform of $f(t)$, and label it $g(\omega)$ as in Eq. (20.10), then Eq. (20.22) can be rewritten

$$f(t) = \sqrt{\frac{1}{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{-i\omega t} d\omega, \quad (20.23)$$

showing that whenever we have the Fourier transform of a function $f(t)$ we can use it to make a **Fourier integral representation** of that function.

The Fourier integral formula, written as in Eq. (20.23), illustrates the value of Fourier analysis in signal processing. If $f(t)$ is an arbitrary signal, Eq. (20.23) describes the signal as composed of a superposition of waves $e^{-i\omega t}$ at angular frequencies¹ ω , with respective amplitudes $g(\omega)$. Thus, the Fourier integral is the underlying justification that one can express a signal either by its time dependence $f(t)$ or by its (angular) frequency distribution $g(\omega)$.

Before leaving the Fourier integral, we should remark that our derivation of it did not provide a rigorous justification for the reversal of the order of integration and the passage to the infinite- n limit. The interested reader can find a more rigorous treatment in, for example, the work *Fourier Transforms* by I. N. Sneddon (Additional Readings).

Example 20.2.3 FOURIER INTEGRAL REPRESENTATION

From the first transform of Example 20.2.1, we found that $f(t) = e^{-\alpha|t|}$ has Fourier transform $g(\omega) = \sqrt{1/2\pi} \, 2\alpha/(\alpha^2 + \omega^2)$. If we substitute these data into Eq. (20.23), we obtain

$$e^{-\alpha|t|} = f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\alpha e^{-i\omega t}}{\alpha^2 + \omega^2} d\omega = \frac{\alpha}{\pi} \int_{-\infty}^{\infty} \frac{e^{-i\omega t}}{\alpha^2 + \omega^2} d\omega. \quad (20.24)$$

Equation (20.24) provides an integral representation for $\exp(-\alpha|t|)$ that contains no absolute value signs and may constitute a useful starting point for various analytical manipulations. We will shortly encounter some more substantive examples with immediate applications for physics. ■

Inverse Fourier Transform

As the reader may have noticed, Eq. (20.23) is a formula for the **inverse Fourier transform**. Note that the regular (“direct”) and inverse Fourier transforms are given by very similar (but not quite identical) formulas. The only difference is in the sign of the complex exponential. This change of sign causes two successive applications of the Fourier transform not to be identical with applying the transform and then its inverse, and the difference shows up when $g(\omega)$ is not real.²

¹The wave $e^{-i\omega t}$ has period $2\pi/\omega$, thus frequency $\nu = \omega/2\pi$. Its angular frequency (radians per unit time rather than cycles) is $2\pi\nu = \omega$.

²Even functions have real Fourier transforms; the transforms of odd functions are imaginary. A function that is neither even nor odd will have a Fourier transform that is complex.

The analysis of the preceding subsection can also be applied to the Fourier cosine and sine transforms. For convenience, we summarize the formulas for all three varieties of the Fourier transform and their respective inverses.

$$g(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt, \quad (20.25)$$

$$f(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(\omega) e^{-i\omega t} d\omega, \quad (20.26)$$

$$g_c(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \cos \omega t dt, \quad (20.27)$$

$$f_c(t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(\omega) \cos \omega t d\omega, \quad (20.28)$$

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} f(t) \sin \omega t dt, \quad (20.29)$$

$$f_s(t) = \sqrt{\frac{2}{\pi}} \int_0^{\infty} g(\omega) \sin \omega t d\omega. \quad (20.30)$$

Note that the Fourier sine and cosine transforms only use data for $0 \leq t < \infty$. Therefore, even though it is possible to evaluate the corresponding inverse transforms for negative t , the results may be irrelevant to the actual situation at those t values. But if our function $f(t)$ is even, then the cosine transform will reproduce it faithfully for negative t ; odd functions will be properly described for negative t by the sine transform.

Example 20.2.4 FINITE WAVE TRAIN

An important application of the Fourier transform is the resolution of a finite pulse into sinusoidal waves. Imagine that an infinite wave train $\sin \omega_0 t$ is clipped by Kerr cell or saturable dye cell shutters so that we have

$$f(t) = \begin{cases} \sin \omega_0 t, & |t| < \frac{N\pi}{\omega_0}, \\ 0, & |t| > \frac{N\pi}{\omega_0}. \end{cases} \quad (20.31)$$

This corresponds to N cycles of our original wave train (Fig. 20.4). Since $f(t)$ is odd, we use the Fourier sine transform, Eq. (20.28), to obtain

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \int_0^{N\pi/\omega_0} \sin \omega_0 t \sin \omega t \, dt. \quad (20.32)$$

Integrating, we find our amplitude function:

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \left[\frac{\sin[(\omega_0 - \omega)(N\pi/\omega_0)]}{2(\omega_0 - \omega)} - \frac{\sin[(\omega_0 + \omega)(N\pi/\omega_0)]}{2(\omega_0 + \omega)} \right]. \quad (20.33)$$

It is of considerable interest to see how $g_s(\omega)$ depends on frequency. For large ω_0 and $\omega \approx \omega_0$, only the first term will be of any importance because of the denominators. It is plotted in Fig. 20.5. This is the amplitude curve for the single-slit diffraction pattern. It has zeros at

$$\frac{\omega_0 - \omega}{\omega_0} = \frac{\Delta\omega}{\omega_0} = \pm \frac{1}{N}, \pm \frac{2}{N}, \quad \text{and so on.} \quad (20.34)$$

For large N , $g_s(\omega)$ may also be interpreted as proportional to a Dirac delta distribution.

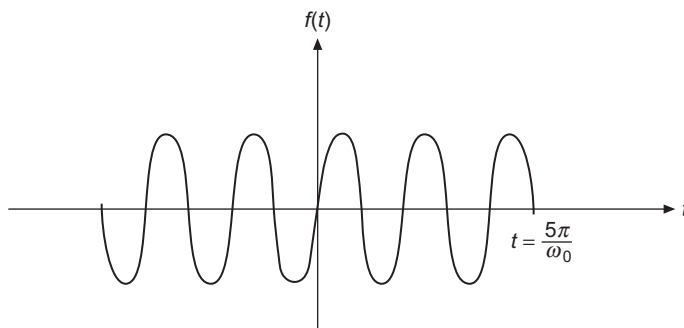


FIGURE 20.4 Finite wave train.

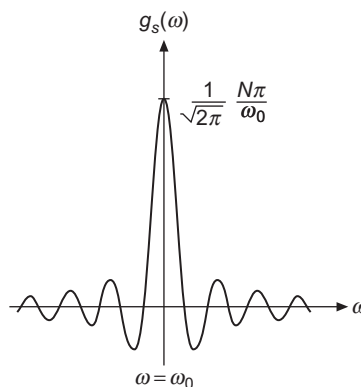


FIGURE 20.5 Fourier transform of finite wave train.

Since a large fraction of the frequency distribution falls within its central maximum, the half-width of that maximum,

$$\Delta\omega = \frac{\omega_0}{N}, \quad (20.35)$$

is a good measure of the spread in angular frequency of our wave pulse. Clearly, if N is large (a long pulse), the frequency spread will be small. On the other hand, if our pulse is clipped short, N small, the frequency distribution will be wider.

The inverse relationship between frequency spread and pulse length is a fundamental property of finite wave distributions; the precision with which a signal can be identified as a specific frequency depends on the pulse length. This same principle finds expression as the **Heisenberg uncertainty principle** of quantum mechanics, in which position uncertainty (the quantum variable corresponding to pulse length) is inversely related to momentum uncertainty (quantum analog of frequency). It is worth noting that the uncertainty principle in quantum mechanics is a consequence of the wave nature of matter and does not depend on additional *ad hoc* postulates. ■

Transforms in 3-D Space

Applying the Fourier transform operator in each dimension of a three-dimensional (3-D) space, we obtain the extremely useful formulas

$$g(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int f(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}} d^3r, \quad (20.36)$$

$$f(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int g(\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{r}} d^3k. \quad (20.37)$$

These integrals are over all space. Verification, if desired, follows immediately by substituting the left-hand side of one equation into the integrand of the other equation and choosing the integration order that permits the complex exponentials to be identified as delta functions in each of the three dimensions. Equation (20.37) may be interpreted as an expansion of a function $f(\mathbf{r})$ in a continuum of plane waves; $g(\mathbf{k})$ then becomes the amplitude of the wave $\exp(-i\mathbf{k}\cdot\mathbf{r})$.

Example 20.2.5 SOME 3-D TRANSFORMS

1. Let's find the Fourier transform of the Yukawa potential, $e^{-\alpha r}/r$. Using the notation $[\dots]^T$ to denote the Fourier transform of the included object, we seek

$$\left[\frac{e^{-\alpha r}}{r} \right]^T(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int \frac{e^{-\alpha r}}{r} e^{i\mathbf{k}\cdot\mathbf{r}} d^3r. \quad (20.38)$$

Perhaps the simplest way to proceed is to introduce the spherical wave expansion for $\exp(i\mathbf{k} \cdot \mathbf{r})$, Eq. (16.61). Equation (20.38), written in spherical polar coordinates, then assumes the form

$$\left[\frac{e^{-\alpha r}}{r} \right]^T(\mathbf{k}) = \frac{4\pi}{(2\pi)^{3/2}} \int_0^\infty r dr \int d\Omega_r \sum_{lm} i^l e^{-\alpha r} j_l(kr) Y_l^m(\Omega_k)^* Y_l^m(\Omega_r). \quad (20.39)$$

All terms of the angular integration vanish except that with $l = m = 0$. For that term, each Y_0^0 has the constant value $1/\sqrt{4\pi}$, and Eq. (20.39) reduces to

$$\left[\frac{e^{-\alpha r}}{r} \right]^T(\mathbf{k}) = \frac{4\pi}{(2\pi)^{3/2}} \int_0^\infty r e^{-\alpha r} j_0(kr) dr. \quad (20.40)$$

Inserting $j_0(kr) = \sin kr / kr$, the r integration becomes elementary, and we reach

$$\left[\frac{e^{-\alpha r}}{r} \right]^T(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \frac{4\pi}{k^2 + \alpha^2}. \quad (20.41)$$

We wrote Eq. (20.41) as we did to make obvious that if the transform were scaled without the factor $1/(2\pi)^{3/2}$, we would have the well-known result $4\pi/(k^2 + \alpha^2)$.

2. Even more important than the Fourier transform of the Yukawa potential is that of the Coulomb potential, $1/r$. An attempt to evaluate this transform directly leads to convergence problems, but it is easy to evaluate it as the limiting case of the Yukawa potential with $\alpha = 0$. Thus, we have the extremely important result,

$$\left[\frac{1}{r} \right]^T(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \frac{4\pi}{k^2}. \quad (20.42)$$

3. From the relation between the Fourier transform and its inverse, Eq. (20.42) can effectively be inverted to yield

$$\left[\frac{1}{r^2} \right]^T(\mathbf{k}) = \left(\frac{\pi}{2} \right)^{1/2} \frac{1}{k}. \quad (20.43)$$

4. Another useful Fourier transform is that of the hydrogenic $1s$ orbital, which (in unnormalized form) is $\exp(-Zr)$. A simple way to evaluate this transform is to differentiate the transform for the Yukawa potential with respect to its parameter, α in Eq. (20.41). Noting that differentiation with respect to this parameter commutes with the transform operator (which involves integration with respect to other variables), we have

$$-\frac{\partial}{\partial Z} \left[\frac{e^{-Zr}}{r} \right]^T(\mathbf{k}) = \left[e^{-Zr} \right]^T(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \frac{8\pi Z}{(k^2 + Z^2)^2}. \quad (20.44)$$

5. Consider next an arbitrary function whose angular dependence is a spherical harmonic (i.e., an angular momentum eigenfunction). Using spherical polar coordinates, we look at

$$\begin{aligned} \left[f(r) Y_l^m(\Omega_r) \right]^T(\mathbf{k}) &= \frac{1}{(2\pi)^{3/2}} \int_0^\infty f(r) r^2 dr \int d\Omega_r Y_l^m(\Omega_r) e^{i\mathbf{k} \cdot \mathbf{r}} \\ &= \frac{4\pi}{(2\pi)^{3/2}} \int_0^\infty f(r) r^2 dr \int d\Omega_r Y_l^m(\Omega_r) \\ &\quad \times \sum_{l'm'} i^{l'} j_{l'}(kr) Y_{l'}^{m'}(\Omega_k) Y_{l'}^{m'}(\Omega_r)^*, \end{aligned}$$

where we have inserted the spherical wave expansion, Eq. (16.61), for $\exp(i\mathbf{k} \cdot \mathbf{r})$. Because the Y_l^m are orthonormal, the summation reduces to a single term, and we have

$$\left[f(r) Y_l^m(\Omega_r) \right]^T(\mathbf{k}) = \frac{4\pi i^l}{(2\pi)^{3/2}} Y_l^m(\Omega_k) \int_0^\infty f(r) j_l(kr) r^2 dr. \quad (20.45)$$

Equation (20.45) shows that a function with spherical harmonic angular dependence has a transform containing the same spherical harmonic and that the radial dependence of the transform is essentially a Hankel transform. Compare with Eq. (20.8).

6. As a final example, consider the Fourier transform of a 3-D Gaussian. Again using spherical polar coordinates and the spherical wave expansion (a procedure generally applicable for transforms of spherically symmetric functions), we get

$$\left[e^{-ar^2} \right]^T(\mathbf{k}) = \frac{4\pi}{(2\pi)^{3/2}} \int_0^\infty r^2 e^{-ar^2} j_0(kr) dr. \quad (20.46)$$

Using methods similar to those of Example 20.2.2, we find

$$\left[e^{-ar^2} \right]^T(\mathbf{k}) = \frac{1}{(2a)^{3/2}} e^{-k^2/4a}. \quad (20.47)$$

This result could also be obtained using Cartesian coordinates and using the result of Example 20.2.2 in each of the three dimensions. ■

Exercises

- 20.2.1** (a) Show that $g(-\omega) = g^*(\omega)$ is a necessary and sufficient condition for $f(x)$ to be real.
 (b) Show that $g(-\omega) = -g^*(\omega)$ is a necessary and sufficient condition for $f(x)$ to be pure imaginary.

Note. The condition of part (a) is used in the development of the dispersion relations of Section 12.8.

20.2.2 The function

$$f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$$

is a symmetrical finite step function.

(a) Find $g_c(\omega)$, Fourier cosine transform of $f(x)$.

(b) Taking the inverse cosine transform, show that

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin \omega \cos \omega x}{\omega} d\omega.$$

(c) From part (b) show that

$$\int_0^{\infty} \frac{\sin \omega \cos \omega x}{\omega} d\omega = \begin{cases} 0, & |x| > 1, \\ \frac{\pi}{4}, & |x| = 1, \\ \frac{\pi}{2}, & |x| < 1. \end{cases}$$

20.2.3 (a) Show that the Fourier sine and cosine transforms of e^{-at} are

$$g_s(\omega) = \sqrt{\frac{2}{\pi}} \frac{\omega}{\omega^2 + a^2}, \quad g_c(\omega) = \sqrt{\frac{2}{\pi}} \frac{a}{\omega^2 + a^2}.$$

Hint. Each of the transforms can be related to the other by integration by parts.

(b) Show that

$$\int_0^{\infty} \frac{\omega \sin \omega x}{\omega^2 + a^2} d\omega = \frac{\pi}{2} e^{-ax}, \quad x > 0,$$

$$\int_0^{\infty} \frac{\cos \omega x}{\omega^2 + a^2} d\omega = \frac{\pi}{2a} e^{-ax}, \quad x > 0.$$

These results can also be obtained by contour integration (Exercise 11.8.12).

20.2.4 Find the Fourier transform of the triangular pulse (Fig. 20.6),

$$f(x) = \begin{cases} h(1 - a|x|), & |x| < 1/a, \\ 0, & |x| > 1/a. \end{cases}$$

Note. This function provides another delta sequence with $h = a$ and $a \rightarrow \infty$.

20.2.5 Consider the sequence

$$\delta_n(x) = \begin{cases} n, & |x| < 1/2n, \\ 0, & |x| > 1/2n. \end{cases}$$

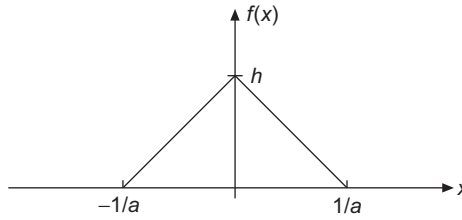


FIGURE 20.6 Triangular pulse.

This is Eq. (1.152). Express $\delta_n(x)$ as a Fourier integral (via the Fourier integral theorem, inverse transform, etc.). Finally, show that we may write

$$\delta(x) = \lim_{n \rightarrow \infty} \delta_n(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} dk.$$

20.2.6 Using the sequence

$$\delta_n(x) = \frac{n}{\sqrt{\pi}} \exp(-n^2 x^2),$$

show that

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} dk.$$

Hint. Remember that $\delta(x)$ is defined in terms of its behavior as part of an integrand.

20.2.7 The formula

$$\delta(t - x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega(t-x)} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} e^{-i\omega x} d\omega$$

can be identified as the continuum limit of an eigenfunction expansion. Derive sine and cosine representations of $\delta(t - x)$ that are comparable to the exponential representation just given.

$$\text{ANS. } \frac{2}{\pi} \int_0^{\infty} \sin \omega t \sin \omega x d\omega, \quad \frac{2}{\pi} \int_0^{\infty} \cos \omega t \cos \omega x d\omega.$$

20.2.8 In a resonant cavity, an electromagnetic oscillation of frequency ω_0 dies out as

$$A(t) = \begin{cases} A_0 e^{-\omega_0 t/2Q} e^{-i\omega_0 t}, & t > 0, \\ 0, & t < 0. \end{cases}$$

The parameter Q is a measure of the ratio of stored energy to energy loss per cycle. Calculate the frequency distribution of the oscillation, $a^*(\omega)a(\omega)$, where $a(\omega)$ is the Fourier transform of $A(t)$.

Note. The larger Q is, the sharper your resonance line will be.

$$\text{ANS. } a^*(\omega)a(\omega) = \frac{A_0^2}{2\pi} \frac{1}{(\omega - \omega_0)^2 + (\omega_0/2Q)^2}.$$

20.2.9 Prove that

$$\frac{\hbar}{2\pi i} \int_{-\infty}^{\infty} \frac{e^{-i\omega t} d\omega}{E_0 - i\Gamma/2 - \hbar\omega} = \begin{cases} \exp\left(-\frac{\Gamma t}{2\hbar}\right) \exp\left(-i\frac{E_0 t}{\hbar}\right), & t > 0, \\ 0, & t < 0. \end{cases}$$

This Fourier integral appears in a variety of problems in quantum mechanics: barrier penetration, scattering, time-dependent perturbation theory, and so on.

Hint. Try contour integration.

20.2.10 Verify that the following are Fourier integral transforms of one another:

$$(a) \quad \left\{ \begin{aligned} &\sqrt{\frac{2}{\pi}} \cdot \frac{1}{\sqrt{a^2 - x^2}}, & |x| < a, \\ &0, & |x| > a, \end{aligned} \right\} \text{ and } J_0(ay),$$

$$(b) \quad \left\{ \begin{aligned} &0, & |x| < a, \\ &-\sqrt{\frac{2}{\pi}} \frac{1}{\sqrt{x^2 + a^2}}, & |x| > a, \end{aligned} \right\} \text{ and } Y_0(a|y|),$$

$$(c) \quad \sqrt{\frac{\pi}{2}} \frac{1}{\sqrt{x^2 + a^2}} \quad \text{and} \quad K_0(a|y|).$$

(d) Can you suggest why $I_0(ay)$ is not included in this list?

Hint. J_0 , Y_0 , and K_0 may be transformed most easily by using an exponential representation, reversing the order of integration, and employing the Dirac delta function exponential representation, Eq. (20.20). These cases can be treated equally well as Fourier cosine transforms.

20.2.11 Show that the following are Fourier transforms of each other:

$$i^n J_n(t) \quad \text{and} \quad \begin{cases} \sqrt{\frac{2}{\pi}} T_n(x)(1-x^2)^{-1/2}, & |x| < 1, \\ 0, & |x| > 1. \end{cases}$$

$T_n(x)$ is the n th-order Chebyshev polynomial.

Hint. With $T_n(\cos \theta) = \cos n\theta$, the transform of $T_n(x)(1-x^2)^{-1/2}$ leads to an integral representation of $J_n(t)$.

20.2.12 Show that the Fourier exponential transform of

$$f(\mu) = \begin{cases} P_n(\mu), & |\mu| \leq 1 \\ 0, & |\mu| > 1 \end{cases}$$

is $(2i^n/2\pi)j_n(kr)$. Here $P_n(\mu)$ is a Legendre polynomial and $j_n(kr)$ is a spherical Bessel function.

20.2.13 (a) Show that $f(x) = x^{-1/2}$ is a **self-reciprocal** under both Fourier cosine and sine transforms; that is,

$$\sqrt{\frac{2}{\pi}} \int_0^{\infty} x^{-1/2} \cos xt \, dx = t^{-1/2},$$

$$\sqrt{\frac{2}{\pi}} \int_0^{\infty} x^{-1/2} \sin xt \, ds = t^{-1/2}.$$

(b) Use the preceding results to evaluate the Fresnel integrals

$$\int_0^{\infty} \cos(y^2) dy \quad \text{and} \quad \int_0^{\infty} \sin(y^2) dy.$$

20.2.14 Show that $\left[\frac{1}{r^2}\right]^T(\mathbf{k}) = \left(\frac{\pi}{2}\right)^{1/2} \frac{1}{k}$.

20.2.15 The Fourier transform formulas for a function of two variables are

$$F(u, v) = \frac{1}{2\pi} \iint f(x, y) e^{i(ux+vy)} \, dx \, dy,$$

$$f(x, y) = \frac{1}{2\pi} \iint F(u, v) e^{-i(ux+vy)} \, du \, dv,$$

where the integrations are over the entire xy or uv plane. For $f(x, y) = f([x^2 + y^2]^{1/2}) = f(r)$, show that the zero-order Hankel transforms

$$F(\rho) = \int_0^{\infty} r f(r) J_0(\rho r) dr,$$

$$f(r) = \int_0^{\infty} \rho F(\rho) J_0(\rho r) d\rho,$$

are a special case of the Fourier transforms.

Note. This technique may be generalized to derive the Hankel transforms of order $\nu = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$. See the two texts by Sneddon (Additional Readings). It might also be noted that the Hankel transforms of half-integral orders $\nu = \pm \frac{1}{2}$ reduce to Fourier sine and cosine transforms.

- 20.2.16** Show that the 3-D Fourier exponential transform of a radially symmetric function may be rewritten as a Fourier sine transform:

$$\frac{1}{(2\pi)^{3/2}} \int_{-\infty}^{\infty} f(r) e^{i\mathbf{k}\cdot\mathbf{r}} d^3x = \frac{1}{k} \sqrt{\frac{2}{\pi}} \int_0^{\infty} r f(r) \sin kr dr.$$

20.3 PROPERTIES OF FOURIER TRANSFORMS

Fourier transforms have a number of useful properties, many of which follow directly from the transform definition. Using the 3-D transform as an illustration, and letting $g(\mathbf{k})$ be the Fourier transform of $f(\mathbf{r})$:

$$\left[f(\mathbf{r} - \mathbf{R}) \right]^T(\mathbf{k}) = e^{i\mathbf{k}\cdot\mathbf{R}} g(\mathbf{k}), \quad (\text{translation}), \quad (20.48)$$

$$\left[f(\alpha\mathbf{r}) \right]^T(\mathbf{k}) = \frac{1}{\alpha^3} g(\alpha^{-1}\mathbf{k}), \quad (\text{change of scale}), \quad (20.49)$$

$$\left[f(-\mathbf{r}) \right]^T(\mathbf{k}) = g(-\mathbf{k}), \quad (\text{sign change}), \quad (20.50)$$

$$\left[f^*(-\mathbf{r}) \right]^T(\mathbf{k}) = g^*(\mathbf{k}), \quad (\text{complex conjugation}), \quad (20.51)$$

$$\left[\nabla f(\mathbf{r}) \right]^T(\mathbf{k}) = -i\mathbf{k} g(\mathbf{k}), \quad (\text{gradient}), \quad (20.52)$$

$$\left[\nabla^2 f(\mathbf{r}) \right]^T(\mathbf{k}) = -k^2 g(\mathbf{k}), \quad (\text{Laplacian}). \quad (20.53)$$

The first four of the above formulas can be obtained by carrying out appropriate operations on the defining equation of the transform; details are left to the exercises. [Equations \(20.52\)](#) and [\(20.53\)](#) are easily established from the inverse transform formula. For example, from [Eq. \(20.37\)](#),

$$\begin{aligned} \nabla f(\mathbf{r}) &= \frac{1}{(2\pi)^{3/2}} \int g(\mathbf{k}) \left[\nabla_r e^{-i\mathbf{k}\cdot\mathbf{r}} \right] d\mathbf{k} \\ &= \frac{1}{(2\pi)^{3/2}} \int g(\mathbf{k}) \left[(-i\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{r}} \right] d\mathbf{k} \\ &= \frac{1}{(2\pi)^{3/2}} \int \left[-i\mathbf{k} g(\mathbf{k}) \right] e^{-i\mathbf{k}\cdot\mathbf{r}} d\mathbf{k}, \end{aligned} \quad (20.54)$$

showing that $-i\mathbf{k} g(\mathbf{k})$ is indeed the Fourier transform of $\nabla f(\mathbf{r})$. It should be noted that this demonstration requires the existence of the integrals involved.

The translation formula is of considerable practical value, as it enables a function that is most conveniently described relative to an origin at $\mathbf{R}$ to have a transform whose natural representation is about the origin in the $\mathbf{k}$ space, albeit with a complex phase factor, $\exp(i\mathbf{k}\cdot\mathbf{R})$. This feature will become important, for example, in problems involving atoms centered at different spatial points, because the transforms of atomic orbitals on such atoms can all be written as centered at a single point in the transform space. Thus, the translation

formula can convert a spatially complex problem into a single-center problem (though now with oscillatory character due to the phase factors).

The formulas for the gradient and Laplacian, as well as their one-dimensional (1-D) variants,

$$\left[f'(t) \right]^T (\omega) = -i\omega g(\omega), \quad (\text{first derivative}), \quad (20.55)$$

$$\left[\frac{d^n}{dt^n} f(t) \right]^T (\omega) = (-i\omega)^n g(\omega), \quad (nth \text{ derivative}), \quad (20.56)$$

make the application of these differential operators have simple forms in the transform space. As we see from Eq. (20.55), the operation of differentiation corresponds in the transform space to multiplication by $-i\omega$.

Example 20.3.1 WAVE EQUATION

Fourier transform techniques may be used to advantage in handling partial differential equations (PDEs). To illustrate the technique, let us derive a familiar expression from elementary physics. An infinitely long string is vibrating freely. The amplitude y of the (small) vibrations satisfies the wave equation

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 y}{\partial t^2}, \quad (20.57)$$

where v is the phase velocity of the wave propagation. We take as initial conditions

$$y(x, 0) = f(x), \quad \left. \frac{\partial y(x, t)}{\partial t} \right|_{t=0} = 0, \quad (20.58)$$

where f is assumed localized, meaning that $\lim_{x \rightarrow \pm\infty} f(x) = 0$.

Our method for solving the PDE of Eq. (20.57) will be to take the Fourier transforms (in x) of its two members, using α as the transform variable. This is equivalent to multiplying Eq. (20.57) by $e^{i\alpha x}$ and integrating over x . Before simplifying, we have

$$\int_{-\infty}^{\infty} \frac{\partial^2 y(x, t)}{\partial x^2} e^{i\alpha x} dx = \frac{1}{v^2} \int_{-\infty}^{\infty} \frac{\partial^2 y(x, t)}{\partial t^2} e^{i\alpha x} dx. \quad (20.59)$$

If we recognize

$$Y(\alpha, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} y(x, t) e^{i\alpha x} dx \quad (20.60)$$

as the transform (from our initial variable x to our transform variable α) of the solution $y(x, t)$ of our PDE, we can rewrite Eq. (20.59) as

$$(-i\alpha)^2 Y(\alpha, t) = \frac{1}{v^2} \frac{\partial^2 Y(\alpha, t)}{\partial t^2}. \quad (20.61)$$

Here we have used Eq. (20.56) for the transform of $\partial^2 y / \partial x^2$ and moved the operator $\partial^2 / \partial t^2$, which is irrelevant to the transform operator, outside the integral, leaving behind just $Y(\alpha, t)$.

Our original problem has now been converted into Eq. (20.61), but this new equation has the important simplifying feature that the only derivative appearing in it is that with respect to t ; we have therefore succeeded in replacing our original PDE (in x and t) with an ordinary differential equation (ODE) (in t only). The dependence of our problem on α (the variable to which x was converted) is only algebraic.

This transformation, from a PDE to an ODE, is a significant achievement. We are now ready to solve Eq. (20.61), subject to the initial conditions, which we need to express in terms of Y . Taking transforms of the quantities in Eq. (20.58), we have

$$\begin{cases} Y(\alpha, 0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{i\alpha x} dx = F(\alpha), \\ \left. \frac{\partial Y(\alpha, t)}{\partial t} \right|_{t=0} = 0. \end{cases} \quad (20.62)$$

It is important to recognize that $F(\alpha)$ is (in principle) known; it is the Fourier transform of the known initial amplitude $f(x)$.

Solving Eq. (20.61) subject to the initial conditions on Y given in Eq. (20.62), we obtain

$$Y(\alpha, t) = F(\alpha) \frac{e^{i\alpha vt} + e^{-i\alpha vt}}{2}. \quad (20.63)$$

We could have written the t dependence as $\cos(\alpha vt)$, but the exponential form is better suited to what we will do next.

Since we really want our solution in terms of x rather than α , our final step will be to apply inverse Fourier transforms to both sides of Eq. (20.63):

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} Y(\alpha, t) e^{-i\alpha x} d\alpha = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(\alpha) \frac{e^{i\alpha vt - i\alpha x} + e^{-i\alpha vt - i\alpha x}}{2} d\alpha. \quad (20.64)$$

The left-hand side of Eq. (20.64) is clearly $y(x, t)$; each term on the right-hand side is an inverse transform of F (and is therefore f), but the first exponential, if written $e^{-i\alpha(x-vt)}$, can be seen to lead to an inverse transform of argument $x - vt$, while the second exponential leads to an inverse transform of argument $x + vt$. Thus, our final simplification of Eq. (20.64) takes the form

$$y(x, t) = \frac{1}{2} \left[f(x - vt) + f(x + vt) \right]. \quad (20.65)$$

Our solution thus consists of a superposition in which half the amplitude of the original wave form is moving toward $+x$ (at velocity v) while the other half of the original wave form is moving (also at velocity v) in the $-x$ direction. ■

Example 20.3.2 HEAT FLOW PDE

To illustrate another transformation of a PDE into an ODE, let us Fourier transform the 1-D heat-flow PDE,

$$\frac{\partial \psi}{\partial t} = a^2 \frac{\partial^2 \psi}{\partial x^2},$$

where the solution $\psi(x, t)$ is the temperature at position x and time t .

We transform the x dependence, with the transform variable denoted y , writing the transform of $\psi(x, t)$ as $\Psi(y, t)$, and identifying the transform of $\partial^2 \psi(x, t)/\partial x^2$ as $-y^2 \Psi(y, t)$. Our heat flow equation then takes the form

$$\frac{\partial \Psi(y, t)}{\partial t} = -a^2 y^2 \Psi(y, t),$$

with general solution

$$\ln \Psi(y, t) = -a^2 y^2 t + \ln C(y), \quad \text{or} \quad \Psi = C(y) e^{-a^2 y^2 t}.$$

The physical significance of $C(y)$ is that it is the initial spatial distribution of Ψ or, in other words, the Fourier transform of the initial temperature profile $\psi(x, 0)$. Thus, if we assume the initial temperature distribution is known, then so also is $C(y)$, and our PDE solution, the inverse transform of Ψ , assumes the form

$$\psi(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} C(y) e^{-a^2 y^2 t} e^{-iyx} dy. \quad (20.66)$$

Further progress depends on the specific form of $C(y)$. If we assume the initial temperature to be a delta-function spike at $x = 0$, corresponding to an instantaneous pulse of thermal energy at $x = t = 0$, we then have as its Fourier transform $C(y) = \text{constant}$, see Eq. (20.14). We can now evaluate the integral in Eq. (20.66) to obtain an explicit form for $\psi(x, t)$. With C constant, the functional form of Eq. (20.66) is (apart from the sign of i) just that encountered in Example 20.2.2 for the Fourier transform of a Gaussian, and we can evaluate the integral to obtain

$$\psi(x, t) = \frac{C}{a\sqrt{2t}} \exp\left(-\frac{x^2}{4a^2 t}\right).$$

This form for ψ was obtained in Section 9.7, but it arose there as a clever guess that was ultimately justified because it led to a solution of the diffusion PDE. ■

Example 20.3.3 COULOMB GREEN'S FUNCTION

The Green's function associated with the Poisson equation satisfies the PDE

$$\nabla_r^2 G(\mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}'). \quad (20.67)$$

We take the Fourier transform of both sides of this equation with respect to $\mathbf{r}$, designating $g(\mathbf{k}, \mathbf{r}')$ as the transform of G . Note that $\mathbf{r}'$ is unaffected by the transformation.

Using Eq. (20.53), the left-hand side of Eq. (20.67) becomes $-k^2 g(\mathbf{k}, \mathbf{r}')$, while the right-hand side, in which the delta function has been translated an amount $\mathbf{r}'$, has according to Eq. (20.48) the transform $e^{i\mathbf{k}\cdot\mathbf{r}'} \delta^T(\mathbf{k})$. Thus, Eq. (20.67) transforms into

$$-k^2 g(\mathbf{k}, \mathbf{r}') = \frac{1}{(2\pi)^{3/2}} e^{i\mathbf{k}\cdot\mathbf{r}'},$$

where the transform of the delta function has been evaluated as the 3-D equivalent of Eq. (20.14). We may now solve for g :

$$g(\mathbf{k}, \mathbf{r}') = -\frac{1}{(2\pi)^{3/2}} \frac{e^{i\mathbf{k}\cdot\mathbf{r}'}}{k^2},$$

and recover G by taking the inverse transform,

$$G(\mathbf{r}, \mathbf{r}') = -\frac{1}{(2\pi)^3} \int \frac{e^{i\mathbf{k}\cdot\mathbf{r}'}}{k^2} e^{-i\mathbf{k}\cdot\mathbf{r}} d^3k = -\frac{1}{(2\pi)^3} \int \frac{d^3k}{k^2} e^{-i\mathbf{k}\cdot(\mathbf{r}-\mathbf{r}')}.$$

We see that the evaluation is proportional to that of the inverse transform of $1/k^2$, but for argument $\mathbf{r} - \mathbf{r}'$. Using Eq. (20.43) (which applies also for the inverse transform because it is real), we reach

$$G(\mathbf{r}, \mathbf{r}') = -\frac{1}{(2\pi)^{3/2}} \left(\frac{\pi}{2}\right)^{1/2} \frac{1}{|\mathbf{r} - \mathbf{r}'|} = -\frac{1}{4\pi} \frac{1}{|\mathbf{r} - \mathbf{r}'|},$$

a result we have previously obtained by other methods (cf. Section 10.2). Note that we did not assume G to be a function of $\mathbf{r} - \mathbf{r}'$; we **found** it to have that form. ■

Successes and Limitations

Some of the above examples illustrate an important role played by the Fourier transform:

- *Use of the Fourier transform can convert a PDE into an ODE, thereby reducing the “degree of transcendence” of the problem.*

All the examples also illustrate the procedure sketched schematically in Fig. 20.1:

- *Fourier transformation can often convert a difficult problem into one which we are able to solve. A useful form for our solution can then be obtained by transforming it back to physical space.*

Despite these successes, it is worth noting that not all problems posed as differential equations are amenable to Fourier-transform solution methods. Some of the limitations arise from the implicit requirement that the necessary transforms and their inverses exist. We can also expect Fourier methods to work only when the solution is unique, as the process of taking a transform and then solving an algebraic equation produces a single result, and not a set of two or more linearly independent solutions.

Usually the boundary conditions are the proximate reason that a differential equation solution is unique, and the requirement that an (exponential) Fourier transform exist

imposes Dirichlet boundary conditions at infinity. For 1-D systems on the semi-infinite range $0 \leq x < \infty$, use of the Fourier sine transform imposes a Dirichlet condition at the finite boundary $x = 0$, while use of the cosine transform corresponds to a Neumann boundary condition there.

Additional opportunities for solving differential equations by transform methods are provided by use of the Laplace transform, for which it is more natural to introduce boundary data. See the later sections of this chapter.

Exercises

- 20.3.1** Write the 1-D equivalents of the equations for translation, scale change, sign change, and complex conjugation that were given for 3-D transforms in Eqs. (20.48) to (20.51).
- 20.3.2** (a) Show that by replacement of $\mathbf{r}$ by $\mathbf{r} - \mathbf{R}$ in the formula for the Fourier transform of $f(\mathbf{r})$, one can derive the translation formula, Eq. (20.48).
 (b) Using methods similar to those for part (a), establish the formulas for scale change, sign change, and complex conjugation, Eqs. (20.49) to (20.51).
- 20.3.3** Derive Eq. (20.53), the formula for the Fourier transform of $\nabla^2 f(\mathbf{r})$.
- 20.3.4** Verify Eqs. (20.55) and (20.56), the formulas for the derivatives of 1-D Fourier transforms.
- 20.3.5** Derive the inverse of Eq. (20.56), namely that

$$[t^n f(t)]^T(\omega) = i^{-n} \frac{d^n}{d\omega^n} g(\omega).$$

- 20.3.6** The 1-D neutron diffusion equation with a (plane) source is

$$-D \frac{d^2 \varphi(x)}{dx^2} + K^2 D \varphi(x) = Q \delta(x),$$

where $\varphi(x)$ is the neutron flux, $Q \delta(x)$ is the (plane) source at $x = 0$, and D and K^2 are constants. Apply a Fourier transform. Solve the equation in transform space. Transform your solution back into x -space.

$$\text{ANS. } \varphi(x) = \frac{Q}{2KD} e^{-|Kx|}.$$

20.4 FOURIER CONVOLUTION THEOREM

An important relationship satisfied by Fourier transforms is that known as the **convolution theorem**. As we shall soon see, this theorem is useful in the solution of differential equations, in establishing the normalization of momentum wave functions, in the evaluation of integrals arising in many branches of physics, and in a variety of signal-processing applications.

We define the **convolution** of two functions $f(x)$ and $g(x)$, understood here to be over the interval $(-\infty, \infty)$, as the following operation designated $f * g$:

$$(f * g)(x) \equiv \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(y) f(x - y) dy. \quad (20.68)$$

The corresponding definition in three dimensions is

$$(f * g)(\mathbf{r}) \equiv \frac{1}{(2\pi)^{3/2}} \int g(\mathbf{r}') f(\mathbf{r} - \mathbf{r}') d^3 r', \quad (20.69)$$

where the integral is over the full 3-D space.

This operation is sometimes referred to as **Faltung**, the German term for “folding.” To better understand the origin of this name, look at [Fig. 20.7](#), where we have plotted $f(y) = e^{-y}$ and $f(x - y) = e^{-(x-y)}$. Clearly, $f(y)$ and $f(x - y)$ are related by reflection relative to the vertical line $y = x/2$; that is, we could generate $f(x - y)$ by folding over $f(y)$ on the line $y = x/2$.

Our interest here is not primarily in the nomenclature, but rather to understand what happens if we take the Fourier transform of a convolution. Letting $F(t)$ and $G(t)$, respectively, be the Fourier transforms of f and g , we find

$$\begin{aligned} (f * g)^T(t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dy g(y) f(x - y) \right] e^{itx} \\ &= \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dy g(y) e^{ity} \right] \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dx f(x - y) e^{it(x-y)} \right] \\ &= \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dy g(y) e^{ity} \right] \left[\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dz f(z) e^{itz} \right] \\ &= G(t) F(t). \end{aligned} \quad (20.70)$$

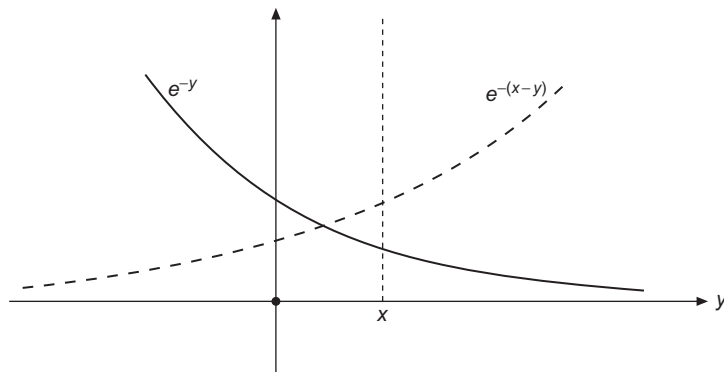


FIGURE 20.7 Factors in a Faltung.

In the second line of the above equation set we simply divided e^{itx} into the two factors e^{ity} and $e^{it(x-y)}$; the third line was reached by changing the integration variable of the second integral from x to $z = x - y$. After this change, y only appears in the first set of square brackets and z only appears in the second bracket set. We are then able to continue to the fourth line where we identify the integrals as Fourier transforms.

We often encounter integrals that have the form of a convolution $f * g$. The convolution theorem then enables the construction of the Fourier transform of the integral, and the integral itself will then be given by taking the inverse transform of $(f * g)^T$. This process corresponds to

$$\begin{aligned} \int_{-\infty}^{\infty} g(y)f(x-y)dy &= \sqrt{2\pi}(f * g)(x) = \sqrt{2\pi} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (f * g)^T(t)e^{-ixt} dt \\ &= \int_{-\infty}^{\infty} G(t)F(t)e^{-ixt} dt. \end{aligned} \quad (20.71)$$

Once again we see an appealing feature inherent to Fourier analysis. While the two functions in our original integral, $g(y)$ and $f(x-y)$, had different arguments, their transforms, $G(t)$ and $F(t)$, have the same argument. We still have an integral to evaluate after using the convolution theorem, but (as just observed) the integrand consists of a product of quantities both of which are evaluated at the **same** point. The cost of the transformation is the presence of a complex exponential, which imparts oscillatory character to the integral. We have thus traded geometric complexity for oscillational complexity. Often this will be an advantageous trade-off.

For the record, here is the 3-D equivalent of Eq. (20.71):

$$\int g(\mathbf{r}')f(\mathbf{r}-\mathbf{r}')d^3r' = \int F(\mathbf{k})G(\mathbf{k})e^{-i\mathbf{k}\cdot\mathbf{r}}d^3k. \quad (20.72)$$

Parseval Relation

If we specialize Eq. (20.71) to $x = 0$, we get the relatively simple result

$$\int_{-\infty}^{\infty} f(-y)g(y)dy = \int_{-\infty}^{\infty} F(t)G(t)dt. \quad (20.73)$$

This equation becomes more easily interpreted if we change $f(y)$ to $f^*(-y)$. Then we must replace $f(-y)$ in Eq. (20.73) by $f^*(y)$, while $F(t)$ becomes $[f^*(-y)]^T$, which, invoking Eq. (20.51), can be written $F^*(t)$. With these changes, we have

$$\int_{-\infty}^{\infty} f^*(y)g(y)dy = \int_{-\infty}^{\infty} F^*(t)G(t)dt. \quad (20.74)$$

This equation is known as the **Parseval relation**; some authors prefer to call it **Rayleigh's theorem**.

The integrals in Eq. (20.74) are of the form of scalar products, and will exist if f and g (and therefore also F and G) are quadratically integrable (i.e., members of an $\mathcal{L}^2$ space). Letting $\mathcal{F}$ denote the Fourier transform operator, we can rewrite Eq. (20.74) in the compact form

$$\langle f|g\rangle = \langle \mathcal{F}f|\mathcal{F}g\rangle. \quad (20.75)$$

If we now move the $\mathcal{F}$ out of the left half-bracket, writing instead its adjoint in the right half-bracket, we reach

$$\langle f|g\rangle = \langle f|\mathcal{F}^\dagger\mathcal{F}g\rangle. \quad (20.76)$$

Since this equation must hold for all f and g in our Hilbert space, it is necessary that $\mathcal{F}^\dagger\mathcal{F}$ reduce to the identity operator, meaning that

$$\mathcal{F}^\dagger = \mathcal{F}^{-1}. \quad (20.77)$$

Our conclusion is that the Fourier transform operator is **unitary**.

If, next, we consider the special case $g = f$, Eq. (20.75) takes the form

$$\langle f|f\rangle = \langle F|F\rangle, \quad (20.78)$$

showing that f and its transform, F , have the same norm, a result that is hardly surprising since we already know that transforming f twice brings us back to at worst f multiplied by a complex phase factor.

An interesting consequence of the unitarity property is illustrated by the formulas governing Fraunhofer diffraction optics. The amplitude of the diffraction pattern appears as the Fourier transform of the function describing the aperture (compare Exercise 20.4.3). With intensity proportional to the square of the amplitude, the Parseval relation implies that the energy passing through the aperture (the integral of $|f|^2$) is equal to that in the diffraction pattern, whose total energy is the integral of $|F|^2$. In this problem the Parseval relation corresponds to energy conservation.

We close this topic with two observations. First, note how the clarity and simplicity of our discussion of the Parseval relation was greatly enhanced by introducing appropriate notation. Much of our insight and intuition regarding mathematical concepts flows directly from the use of good notations for their description. Secondly, we call attention to the fact that Parseval's relation can be developed independently of the inverse Fourier transform and then used rigorously to derive the inverse transform. Details can be found in the text by Morse and Feshbach (Additional Readings).

Here are some examples illustrating use of the convolution theorem.

Example 20.4.1 POTENTIAL OF CHARGE DISTRIBUTION

We require the potential at all points $\mathbf{r}$ produced by a charge distribution $\rho(\mathbf{r}')$. From Coulomb's law, or equivalently from the Green's function for Poisson's equation, we have

$$\psi(\mathbf{r}) = \frac{1}{4\pi} \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r'. \quad (20.79)$$

The integral for ψ is of the convolution form, and its presence in this problem suggests that convolutions will arise in a wide variety of problems in which there is a distributed source of almost any kind and an effect therefrom that depends on relative position.

Taking $f(\mathbf{r}) = 1/r$, so that $f(\mathbf{r} - \mathbf{r}') = 1/|\mathbf{r} - \mathbf{r}'|$, and $g(\mathbf{r}) = \rho(\mathbf{r})$, application of the convolution formula Eq. (20.72) yields

$$\psi(\mathbf{r}) = \frac{1}{4\pi} \int f^T(\mathbf{k}) g^T(\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{r}} d^3k.$$

Since

$$f^T(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \frac{4\pi}{k^2} \quad \text{and} \quad g^T(\mathbf{k}) = \rho^T(\mathbf{k}),$$

we have

$$\psi(\mathbf{r}) = \frac{1}{(2\pi)^{3/2}} \int \frac{\rho^T(\mathbf{k})}{k^2} e^{-i\mathbf{k}\cdot\mathbf{r}} d^3k. \quad (20.80)$$

Depending on the functional form of ρ , Eq. (20.80) may or may not be easier to evaluate than the original equation for ψ , Eq. (20.79). ■

Example 20.4.2 TWO-CENTER OVERLAP INTEGRAL

In quantum mechanics problems involving molecules, one often encounters the so-called **overlap integral**, which is the scalar product of two atomic orbitals, one, φ_a , centered at a point **A**, and another, φ_b , centered at a different point **B**. This overlap integral, denoted S_{ab} , can be written

$$S_{ab} = \int \varphi_a^*(\mathbf{r} - \mathbf{A}) \varphi_b(\mathbf{r} - \mathbf{B}) d^3r. \quad (20.81)$$

The integral is over the full 3-D space. One way to evaluate S_{ab} starts by changing to coordinates in which the origin is at **A**; this amounts to the substitution $\mathbf{r}' = \mathbf{r} - \mathbf{A}$, in terms of which $\mathbf{r} - \mathbf{B} = \mathbf{r}' - (\mathbf{B} - \mathbf{A})$, so

$$S_{ab} = \int \varphi_a^*(\mathbf{r}') \varphi_b(\mathbf{r}' - \mathbf{R}) d^3r',$$

where $\mathbf{R} = \mathbf{B} - \mathbf{A}$. We note the physically expected feature that the value of S_{ab} does not depend on **A** and **B** separately but only on the vector **R** describing their relative position.

This integral for S_{ab} is almost in the standard form for a convolution (it differs therefrom by having $\mathbf{r}' - \mathbf{R}$ instead of $\mathbf{R} - \mathbf{r}'$). This discrepancy can be handled by invoking Eq. (20.50); the net effect is to change the sign of the transform variable **k** when we evaluate φ_b^T .

Again using Eq. (20.72), we write

$$S_{ab} = \int \left[\varphi_a^* \right]^T(\mathbf{k}) \varphi_b^T(-\mathbf{k}) e^{-i\mathbf{k}\cdot\mathbf{R}} d^3k.$$

We continue with the specific case that φ_a and φ_b are Slater-type orbitals (STOs), both with the same screening parameter ζ . These STOs, and their Fourier transforms (which can be obtained by differentiating Eq. (20.41) with respect to its parameter α), are

$$\varphi = \varphi^* = e^{-\zeta r}, \quad \varphi^T = \frac{1}{(2\pi)^{3/2}} \frac{8\pi\zeta}{(k^2 + \zeta^2)^2}.$$

Inserting the formula for φ^T into the integral for S_{ab} , we get

$$S_{ab} = \frac{(8\pi\zeta)^2}{(2\pi)^3} \int \frac{e^{-i\mathbf{k}\cdot\mathbf{R}}}{(k^2 + \zeta^2)^4} d^3k.$$

At this point we already see an advantage of the convolution-based procedure. This integral (whether or not we can easily evaluate it) has assumed a single-center character, with the interorbital spacing relegated to the complex exponential factor.

To complete the evaluation, we now insert the spherical wave expansion for $\exp(-i\mathbf{k}\cdot\mathbf{R})$, Eq. (16.61), and we note the further simplification that the only term surviving the integration over the angular coordinates of $\mathbf{k}$ is the $l = 0$ term of the expansion. Keeping in mind that $Y_0^0 = 1/\sqrt{4\pi}$, that term is seen to be just $j_0(kR)$, so our formula for S_{ab} becomes

$$S_{ab} = \frac{(8\pi\zeta)^2}{(2\pi)^3} \int_0^\infty \frac{j_0(kR)}{(k^2 + \zeta^2)^4} 4\pi k^2 dk.$$

We now have a known 1-D integral, which in fact we encountered in Exercise 14.7.10:

$$k_n(x) = \frac{2^{n+2} (n+1)!}{\pi x^{n+1}} \int_0^\infty \frac{k^2 j_0(kx)}{(k^2 + 1)^{n+2}} dk.$$

Changing x in this formula to ζR and replacing k by k/ζ , we reach

$$S_{ab} = \frac{\pi R^3}{3} k_2(\zeta R) = \frac{\pi e^{-\zeta R}}{3\zeta^3} \left(\zeta^2 R^2 + 3\zeta R + 3 \right). \quad (20.82)$$

Note that when we insert the explicit form for k_2 , we obtain a relatively simple final result.

There are other ways to obtain this formula (one of which is to use prolate ellipsoidal coordinates with $\mathbf{A}$ and $\mathbf{B}$ as foci), but the method we have chosen here provides a good illustration of the issues and formulas that arise when the convolution method is applicable. ■

Multiple Convolutions

Some important problems take the form of multiple convolutions, which we illustrate in one dimension by the convolution of a function h first with a function g followed by the

convolution of that result with f , i.e., $f * (g * h)$. Thus,

$$\begin{aligned} [f * (g * h)](x) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dy f(y) (g * h)(x - y) \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dt f(y) g(t) h(x - y - t), \end{aligned}$$

which after making the substitution $t = z - y$ (and therefore $x - y - t = x - z$) becomes

$$[f * (g * h)](x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz f(y) g(z - y) h(x - z). \quad (20.83)$$

Letting F , G , and H be the Fourier transforms of f , g , and h , this case of the convolution theorem is

$$[f * g * h]^T(\omega) = F(\omega)G(\omega)H(\omega). \quad (20.84)$$

We have now omitted the parentheses surrounding $g * h$ since we would have gotten the same result if we convoluted f , g , and h in any order. Then, taking the inverse transform, we have

$$\int_{-\infty}^{\infty} dy \int_{-\infty}^{\infty} dz f(y) g(z - y) h(x - z) = (2\pi)^{1/2} \int_{-\infty}^{\infty} F(\omega)G(\omega)H(\omega)e^{-i\omega x} d\omega. \quad (20.85)$$

In three dimensions, the corresponding formulas are

$$[f * (g * h)](\mathbf{r}) = \frac{1}{(2\pi)^3} \int d^3 r' \int d^3 r'' f(\mathbf{r}') g(\mathbf{r}'' - \mathbf{r}') h(\mathbf{r} - \mathbf{r}''), \quad (20.86)$$

$$[f * (g * h)]^T(\mathbf{k}) = F(\mathbf{k})G(\mathbf{k})H(\mathbf{k}), \quad (20.87)$$

$$\int d^3 r' \int d^3 r'' f(\mathbf{r}') g(\mathbf{r}'' - \mathbf{r}') h(\mathbf{r} - \mathbf{r}'') = (2\pi)^{3/2} \int F(\mathbf{k})G(\mathbf{k})H(\mathbf{k})e^{-i\mathbf{k}\cdot\mathbf{r}} d^3 k. \quad (20.88)$$

Example 20.4.3 INTERACTION OF TWO CHARGE DISTRIBUTIONS

The electrostatic interaction of two charge distributions $\rho_1(\mathbf{r})$ and $\rho_2(\mathbf{r})$ is given by the integral

$$V = \int d^3 r' \int d^3 r'' \frac{\rho_1(\mathbf{r}')\rho_2(\mathbf{r}'')}{|\mathbf{r}'' - \mathbf{r}'|}, \quad (20.89)$$

which is a double convolution, as in Eq. (20.88), but with the free argument $\mathbf{r}$ set to zero and with a sign discrepancy in the argument of h (which is ρ_2 of the present example).

Taking the above into account and applying Eq. (20.88), we have

$$\begin{aligned} V &= (2\pi)^{3/2} \int d^3k \, \rho_1^T(\mathbf{k}) \left[\frac{1}{r} \right]^T(\mathbf{k}) \rho_2^T(-\mathbf{k}) \\ &= 4\pi \int \frac{d^3k}{k^2} \rho_1^T(\mathbf{k}) \rho_2^T(-\mathbf{k}), \end{aligned} \quad (20.90)$$

where we have inserted the value of $(1/r)^T$ from Eq. (20.42). This expression has the obvious advantage that it is a 3-D integral in place of the original six-fold integration in Eq. (20.89). The price we have to pay for this simplification is the cost of taking the Fourier transforms of ρ_1 and ρ_2 . ■

Transform of a Product

The similarity between the formulas for the direct and inverse Fourier transforms suggest that we may be able to identify the Fourier transform of a product as a convolution. Accordingly, we rewrite Eq. (20.71) with x replaced by $-x$ and also change the variable of integration in that equation from y to $-y$. We then have (multiplying the equation by $1/\sqrt{2\pi}$)

$$\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(-y) f(y-x) dy = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} G(t) F(t) e^{ixt} dt = \left[G(t) F(t) \right]^T(x). \quad (20.91)$$

If we now make the further identifications

$$\left[G(t) \right]^T(y) = g(-y) \quad \text{and} \quad \left[F(t) \right]^T(x-y) = f(y-x),$$

we have

$$(F^T * G^T)(x) = \left[G(t) F(t) \right]^T(x). \quad (20.92)$$

Rewriting Eq. (20.92) with the functions renamed f and g and their respective transforms denoted F and G , we have our desired final result:

$$\left[f g \right]^T = F * G. \quad (20.93)$$

Equation (20.93) will be useful only if f and g individually have Fourier transforms. It is possible that this condition is not satisfied despite the fact that fg possesses a transform. We therefore proceed to consider the case that f not have a transform, but instead possesses a Maclaurin expansion, and therefore can be represented by a series in positive integer powers of x . Then, starting from the relation

$$\left[x^n g(x) \right]^T(t) = i^{-n} \frac{d^n}{dt^n} G(t),$$

the topic of Exercise 20.3.5, we can write

$$\left[f g \right]^T(t) = f \left(-i \frac{d}{dt} \right) G(t), \quad (20.94)$$

where the expression $-i(d/dt)$ is the **argument** of f (and not a multiplicative factor). Unless f is quite simple, this expression may be of limited practical value.

Momentum Space

Hamilton's equations of classical mechanics formalize a symmetry between position variables q and the corresponding (**conjugate**) momentum variables p . This same correspondence carries over into quantum mechanics, where (in one dimension, in units with $\hbar = 1$), the fundamental relationship is the commutator $[x, p] = i$. The time-independent Schrödinger equation (for a particle of mass m) is

$$H\psi \equiv \left[\frac{1}{2m} p^2 + V(x) \right] \psi = E\psi,$$

and it is usually made more explicit by taking $p = -i(d/dx)$, in which case the wave function ψ is a function of x : $\psi = \psi(x)$. In principle we could have chosen p as the fundamental variable, in which case the proper value of the commutator is recovered if we take $x = +i(d/dp)$, and ψ (which we will now give the name φ) will be a function of p : $\varphi = \varphi(p)$. These two representations of the Schrödinger equation in one dimension correspond, respectively, to the two ODEs:

$$-\frac{1}{2m} \frac{d^2}{dx^2} \psi(x) + V(x)\psi(x) = E\psi(x), \quad (20.95)$$

$$\frac{p^2}{2m} \varphi(p) + V\left(i \frac{d}{dp}\right) \varphi(p) = E\varphi(p). \quad (20.96)$$

Note that in the second of these two equations, the argument of V is a differential operator, and unless the form of V is relatively simple, the momentum-space ODE will be quite complicated and correspondingly difficult to solve.

In the **coordinate representation** ($x, -id/dx$), a wave function $\exp(ikx)$ is an eigenfunction of momentum with eigenvalue k :

$$p e^{ikx} = -i \frac{d}{dx} e^{ikx} = -i(ik) e^{ikx} = k e^{ikx},$$

and this fact suggests that momentum wave functions will be Fourier transforms of their coordinate counterparts. We therefore seek to verify the consistency of Eqs. (20.95) and (20.96) by Fourier transforming the first of these two equations, letting $g(t)$ represent the transform of ψ and using Eq. (20.56) to take the transform of the second derivative.³ In the case that V has a Maclaurin expansion, we then use Eq. (20.94), obtaining

$$\frac{t^2}{2m} g(t) + V\left(-i \frac{d}{dt}\right) g(t) = E g(t).$$

This equation can be brought into agreement with Eq. (20.96) if we take its complex conjugate (assuming V to be real), so we can make the identification $\varphi(p) \longleftrightarrow g^*(t)$.

³Here t is the transform variable; in the present context it has nothing to do with time.

On the other hand, if V has a transform we can use the convolution formula, Eq. (20.93), thereby converting Eq. (20.95) into an integral equation:

$$\frac{p^2}{2m} \varphi(p) + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} V^T(p - p') \varphi(p') dp' = E \varphi(p). \quad (20.97)$$

Example 20.4.4 MOMENTUM-SPACE SCHRÖDINGER EQUATION

The time-independent Schrödinger equation for the hydrogen atom has (in **hartree atomic units** $\hbar = m = e = 1$) the coordinate representation

$$-\frac{1}{2} \nabla^2 \psi(\mathbf{r}) - \frac{1}{r} \psi(\mathbf{r}) = E \psi(\mathbf{r}).$$

Taking the Fourier transform of this equation, we get for the **momentum-space** wave function $\varphi(\mathbf{k})$

$$\frac{k^2}{2} \varphi(\mathbf{k}) - \frac{1}{(2\pi)^3} \int \frac{4\pi}{|\mathbf{k} - \mathbf{k}'|^2} \varphi(\mathbf{k}') d^3 k' = E \varphi(\mathbf{k}). \quad (20.98)$$

In reaching Eq. (20.98), we have used the 3-D version of Eq. (20.97), inserting for the transform of V the result from Eq. (20.42).

In principle one can solve Eq. (20.98) for $\varphi(\mathbf{k})$ and the corresponding eigenvalues E , and the results should be equivalent to the original equation. That is a more difficult task than we will undertake now, but it is straightforward to verify that the Fourier transform of the known solution for the hydrogen ground state is a solution to Eq. (20.98).

From Eq. (20.44), the hydrogen $1s$ wave function e^{-r} is seen to have Fourier transform

$$\varphi(\mathbf{k}) = \frac{C}{(k^2 + 1)^2},$$

where C is independent of k and has a value that is irrelevant here. Inserting this result into Eq. (20.98), we find

$$\frac{1}{2} \frac{Ck^2}{(k^2 + 1)^2} - \frac{C}{2\pi^2} \int \frac{d^3 k'}{|\mathbf{k} - \mathbf{k}'|^2 (k'^2 + 1)^2} = E \frac{C}{(k^2 + 1)^2}. \quad (20.99)$$

Writing $|\mathbf{k} - \mathbf{k}'|^2 = k^2 + 2kk' \cos \theta + k'^2$, the integral, though a bit tedious, is found to be elementary. Inserting its value, Eq. (20.99) becomes (canceling the common factor C),

$$\frac{1}{2} \frac{k^2}{(k^2 + 1)^2} - \frac{1}{2} \frac{1}{k^2 + 1} = E \frac{1}{(k^2 + 1)^2}.$$

This equation is satisfied if $E = -1/2$, the correct energy (in hartree atomic units) for the hydrogen $1s$ state. ■

Exercises

20.4.1 Work out the convolution equation corresponding to Eq. (20.71) for

(a) Fourier sine transforms

$$\frac{1}{2} \int_0^{\infty} g(y) [f(y+x) + f(y-x)] dy = \int_0^{\infty} F_s(s) G_s(s) \cos sx \, ds,$$

where f and g are odd functions.

(b) Fourier cosine transforms

$$\frac{1}{2} \int_0^{\infty} g(y) [f(y+x) + f(x-y)] dy = \int_0^{\infty} F_c(s) G_c(s) \cos sx \, ds,$$

where f and g are even functions.

20.4.2 Show that for both Fourier sine and Fourier cosine transforms Parseval's relation has the form

$$\int_0^{\infty} F(t) G(t) dt = \int_0^{\infty} f(y) g(y) dy.$$

20.4.3 (a) A rectangular pulse is described by

$$f(x) = \begin{cases} 1, & |x| < a, \\ 0, & |x| > a. \end{cases}$$

Show that the Fourier exponential transform is

$$F(t) = \sqrt{\frac{2}{\pi}} \frac{\sin at}{t}.$$

This is the single-slit diffraction problem of physical optics. The slit is described by $f(x)$. The diffraction pattern **amplitude** is given by the Fourier transform $F(t)$.

(b) Use the Parseval relation to evaluate

$$\int_{-\infty}^{\infty} \frac{\sin^2 t}{t^2} dt.$$

This integral may also be evaluated by using the calculus of residues (Exercise 11.8.9).

ANS. (b) π .

20.4.4 Solve Poisson's equation, $\nabla^2 \psi(\mathbf{r}) = -\rho(\mathbf{r})/\epsilon_0$, by the following sequence of operations:

- (a) Take the Fourier transform of both sides of this equation. Solve for the Fourier transform of $\psi(\mathbf{r})$.
- (b) Carry out the Fourier inverse transform.

20.4.5 (a) Given $f(x) = 1 - |x/2|$ for $-2 \leq x \leq 2$, with $f(x) = 0$ elsewhere, show that the Fourier transform of $f(x)$ is

$$F(t) = \sqrt{\frac{2}{\pi}} \left(\frac{\sin t}{t} \right)^2.$$

- (b) Using the Parseval relation, evaluate

$$\int_{-\infty}^{\infty} \left(\frac{\sin t}{t} \right)^4 dt.$$

ANS. (b) $\frac{2\pi}{3}$.

20.4.6 With $F(t)$ and $G(t)$ the Fourier transforms of $f(x)$ and $g(x)$, respectively, show that

$$\int_{-\infty}^{\infty} |f(x) - g(x)|^2 dx = \int_{-\infty}^{\infty} |F(t) - G(t)|^2 dt.$$

If $g(x)$ is an approximation to $f(x)$, the preceding relation indicates that the mean square deviation in t -space is equal to the mean square deviation in x -space.

20.4.7 Use the Parseval relation to evaluate

$$(a) \int_{-\infty}^{\infty} \frac{d\omega}{(\omega^2 + a^2)^2}, \quad (b) \int_{-\infty}^{\infty} \frac{\omega^2 d\omega}{(\omega^2 + a^2)^2}.$$

Hint. Compare [Exercise 20.2.3](#).

ANS. (a) $\frac{\pi}{2a^3}$, (b) $\frac{\pi}{2a}$.

20.4.8 The nuclear form factor $F(\mathbf{k})$ and the charge distribution $\rho(\mathbf{r})$ are 3-D Fourier transforms of each other:

$$F(\mathbf{k}) = \frac{1}{(2\pi)^{3/2}} \int \rho(\mathbf{r}) e^{i\mathbf{k}\cdot\mathbf{r}} d^3r.$$

If the measured form factor is

$$F(\mathbf{k}) = (2\pi)^{-3/2} \left(1 + \frac{k^2}{a^2} \right)^{-1},$$

find the corresponding charge distribution.

ANS. $\rho(\mathbf{r}) = \frac{a^2}{4\pi} \frac{e^{-ar}}{r}.$

- 20.4.9** Using convolution methods, find an integral whose value is the electrostatic interaction energy between a charge distribution $\rho(\mathbf{r} - \mathbf{A})$ and a unit point charge at $\mathbf{C}$.
- 20.4.10** With $\psi(\mathbf{r})$ a wave function in ordinary space and $\varphi(\mathbf{p})$ the corresponding momentum function, show that

$$(a) \quad \frac{1}{(2\pi\hbar)^{3/2}} \int \mathbf{r} \psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r = i\hbar \nabla_p \varphi(\mathbf{p}),$$

$$(b) \quad \frac{1}{(2\pi\hbar)^{3/2}} \int \mathbf{r}^2 \psi(\mathbf{r}) e^{-i\mathbf{r}\cdot\mathbf{p}/\hbar} d^3r = (i\hbar \nabla_p)^2 \varphi(\mathbf{p}).$$

Note. ∇_p is the gradient in momentum space:

$$\hat{\mathbf{e}}_x \frac{\partial}{\partial p_x} + \hat{\mathbf{e}}_y \frac{\partial}{\partial p_y} + \hat{\mathbf{e}}_z \frac{\partial}{\partial p_z}.$$

These results may be extended to any positive integer power of $\mathbf{r}$ and therefore to any (analytic) function that may be expanded as a Maclaurin series in $\mathbf{r}$.

- 20.4.11** The ordinary space wave function $\psi(\mathbf{r}, t)$ satisfies the time-dependent Schrödinger equation,

$$i\hbar \frac{\partial \psi(\mathbf{r}, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\mathbf{r}) \psi.$$

Show that the corresponding time-dependent momentum wave function satisfies the analogous equation

$$i\hbar \frac{\partial \varphi(\mathbf{p}, t)}{\partial t} = \frac{p^2}{2m} \varphi + V(i\hbar \nabla_p) \varphi.$$

Note. Assume that $V(\mathbf{r})$ may be expressed by a Maclaurin series and use [Exercise 20.4.10](#). $V(i\hbar \nabla_p)$ is the same function of the variable $i\hbar \nabla_p$ that $V(\mathbf{r})$ is of the variable $\mathbf{r}$.

20.5 SIGNAL-PROCESSING APPLICATIONS

A time-dependent electrical pulse $f(t)$ may be regarded as a superposition of waves of many frequencies. For angular frequency ω , we have a contribution

$$F(\omega) e^{i\omega t}.$$

Then the complete pulse may be written as

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{i\omega t} d\omega. \quad (20.100)$$

Because the angular frequency ω is related to the linear frequency ν by

$$\nu = \frac{\omega}{2\pi},$$

most physicists associate the entire $1/2\pi$ factor with this integral, so this formula differs by a factor $(2\pi)^{-1/2}$ from the definition we have adopted for the Fourier transform.

But if ω is a frequency, what about the negative frequencies? The negative ω may be looked on as a mathematical device to avoid dealing with two functions ($\cos \omega t$ and $\sin \omega t$) separately.

Because Eq. (20.100) has the form of a Fourier transform, we may solve for $F(\omega)$ by taking the inverse transform. Keeping in mind the scale at which we wrote Eq. (20.100), we get

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt. \quad (20.101)$$

Equation (20.101) represents a **resolution of the pulse** $f(t)$ into its angular frequency components. Equation (20.100) is a **synthesis of the pulse** from its components.

Now consider some device, such as a servomechanism or a stereo amplifier, with an input $f(t)$ and an output $g(t)$. For an input of a single frequency f_ω with input $f_\omega(t) = F(\omega)e^{i\omega t}$, the device will alter the amplitude and may also change the phase. For the situations we discuss here, we assume a linear response, which means that we are assuming that g_ω (the output corresponding to f_ω) will be a signal at the same frequency as f_ω , will scale linearly with f_ω , and be independent of the simultaneous presence of signals at other frequencies. However, the responses of interesting devices will depend on the frequency. Hence, our assumption is that g_ω and f_ω are related by an equation of the form

$$g_\omega(t) = \varphi(\omega) f_\omega(t). \quad (20.102)$$

This amplitude- and phase-modifying function, $\varphi(\omega)$, is called a **transfer** function. When making schematic diagrams of electronic circuits, it is customary to designate a device characterized by a transfer function by a suitably labeled box with input and output conductors, as shown in (Fig. 20.8).

Because we have assumed the operation corresponding to the transfer function to be linear, the total output from a pulse containing many frequencies may be obtained by integrating over the entire input, as modified by the transfer function,

$$g(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(\omega) F(\omega) e^{i\omega t} d\omega. \quad (20.103)$$

The transfer function is characteristic of the device to which it applies. Once it is known (either by calculation or measurement), the output $g(t)$ can be calculated for any input $f(t)$.

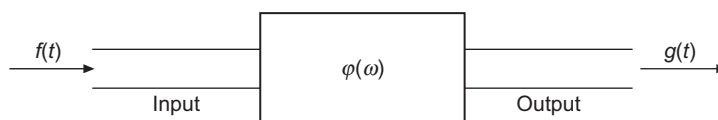


FIGURE 20.8 Schematic for device described by transfer function.

Equation (20.103) can be brought to a convenient form if we recognize that it is simply the formula for the Fourier transform of the product $\varphi(\omega)F(\omega)$. We already know that $F(\omega)$ has transform $f(t)$. Letting $\Phi(t)$ be (at the scaling of this section) the transform of $\varphi(\omega)$, we may then use Eq. (20.93) to rewrite Eq. (20.103) as the convolution of the transforms f and Φ :

$$g(t) = \int_{-\infty}^{\infty} f(t')\Phi(t-t')dt'. \quad (20.104)$$

Interpreting Eq. (20.104), we have an input (a “cause”), namely $f(t')$, modified by $\Phi(t-t')$, producing an output (an “effect”), namely $g(t)$. Adopting the concept of **causality** (that the cause precedes the effect), we must obtain contributions to $g(t)$ only from times t' such that $t' < t$. We do this by requiring

$$\Phi(t-t') = 0, \quad t' > t. \quad (20.105)$$

Then Eq. (20.104) becomes

$$g(t) = \int_{-\infty}^t f(t')\Phi(t-t')dt'. \quad (20.106)$$

Since Eq. (20.106) must yield real output $g(t)$ for arbitrary real input $f(t)$, we see that in addition to the requirement in Eq. (20.105), we also know that $\Phi(t)$ must be real.

The adoption of Eq. (20.106) and the reality of Φ have profound consequences here and equivalently in dispersion theory (Section 12.8).

Example 20.5.1 TRANSFER FUNCTION: HIGH-PASS FILTER

A **high-pass filter** permits almost complete transmission of high-frequency electrical signals but strongly attenuates those at lower frequencies. A very simple high-pass filter is shown in Fig. 20.9. Its transfer function describes the steady-state behavior of the filter in the absence of loading (meaning that the output terminals are not connected to anything), so we can assume that, for a signal at frequency ω , the input, output, and current are the real parts of the respective quantities $V_{\text{in}}e^{i\omega t}$, $V_{\text{out}}e^{i\omega t}$, $Ie^{i\omega t}$. Possible phase differences in these quantities are allowed for by permitting V_{in} , V_{out} , and I to be complex.

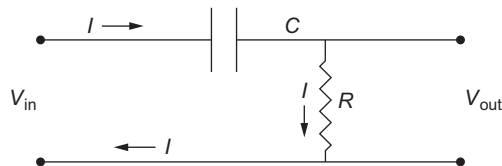


FIGURE 20.9 Simple high-pass filter.

Following the usual procedure for electrical circuit analysis, we solve Kirchhoff's equation (the condition that the net change in potential around any loop of the circuit vanishes):

$$V_{\text{in}} e^{i\omega t} = \int \frac{I}{C} e^{i\omega t} dt + R I e^{i\omega t}. \quad (20.107)$$

Differentiating with respect to t (to eliminate the integral), we have

$$V_{\text{in}} \frac{d}{dt} e^{i\omega t} = \frac{I}{C} e^{i\omega t} + R I \frac{d}{dt} e^{i\omega t},$$

which, evaluating the derivatives, reduces to

$$i\omega V_{\text{in}} = \frac{I}{C} + i\omega R I, \quad \text{with solution} \quad I = \frac{i\omega C V_{\text{in}}}{1 + i\omega R C}. \quad (20.108)$$

Since $V_{\text{out}} = I R$, we easily find the transfer function

$$\varphi(\omega) = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{i\omega R C}{1 + i\omega R C}. \quad (20.109)$$

To confirm the behavior of the filter, note that in the limit of large ω , $\varphi(\omega) \rightarrow 1$, while at small ω , $\varphi(\omega) \rightarrow i\omega R C$, which vanishes in the limit of small ω . The transition between these two limiting behaviors is a function of the product $R C$. ■

Limitations on Transfer Functions

Let us write the transfer function $\varphi(\omega)$ as the inverse Fourier transform of $\Phi(t)$ (still using the scaling of this section), keeping in mind that $\Phi(t)$ vanishes for $t < 0$,

$$\varphi(\omega) = \int_0^{\infty} \Phi(t) e^{-i\omega t} dt. \quad (20.110)$$

Now, separating φ into its real and imaginary parts: $\varphi(\omega) = u(\omega) + i v(\omega)$, and making the same separation for the right-hand side of Eq. (20.110), we have

$$\begin{aligned} u(\omega) &= \int_0^{\infty} \Phi(t) \cos \omega t dt, \\ v(\omega) &= - \int_0^{\infty} \Phi(t) \sin \omega t dt. \end{aligned} \quad (20.111)$$

These formulas tell us that $u(\omega)$ is even, and that $v(\omega)$ is odd.

Since Eqs. (20.111) are cosine and sine transforms, they can be inverted to give two alternative formulas for $\Phi(t)$ in the range of applicability of these transforms, namely for

$t > 0$. Continuing to use the transform scaling of this section,

$$\begin{aligned}\Phi(t) &= \frac{2}{\pi} \int_0^{\infty} u(\omega) \cos \omega t \, d\omega, \\ &\quad (t > 0) \quad (20.112) \\ &= -\frac{2}{\pi} \int_0^{\infty} v(\omega) \sin \omega t \, d\omega.\end{aligned}$$

The present significance of these results is that

$$\int_0^{\infty} u(\omega) \cos \omega t \, d\omega = - \int_0^{\infty} v(\omega) \sin \omega t \, d\omega, \quad (t > 0). \quad (20.113)$$

The imposition of causality has led to a mutual interdependence of the real and imaginary parts of the transfer function. The present result is similar to those involving causality that were discussed in Section 12.8.

We close this subsection by verifying that the conditions on u and v are consistent with the properties required of Φ . Writing

$$\Phi(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \varphi(\omega) e^{i\omega t} \, d\omega,$$

then inserting $e^{i\omega t} = \cos \omega t + i \sin \omega t$ and $\varphi = u + i v$, we have

$$\begin{aligned}\Phi(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[u(\omega) \cos \omega t - v(\omega) \sin \omega t \right] d\omega \\ &\quad + \frac{i}{2\pi} \int_{-\infty}^{\infty} \left[u(\omega) \sin \omega t + v(\omega) \cos \omega t \right] d\omega. \quad (20.114)\end{aligned}$$

The imaginary part of Eq. (20.114) vanishes because its integrand is an odd function of ω . If $t > 0$, we know from Eq. (20.113) that the two terms of the real part of Eq. (20.114) are equal, and we get the expected nonzero result. But if $t < 0$, the sign of the second term of the real part is changed and they then add to zero.

Exercises

20.5.1 Find the transfer function $\varphi(\omega)$ for the circuit shown in the left panel of Fig. 20.10. Is this a high-pass, a low-pass, or a more complicated filter?

20.5.2 Find the transfer function $\varphi(\omega)$ for the circuit shown in the right panel of Fig. 20.11.

Hint. The potential difference across an inductor is given by $L \, dI/dt$.

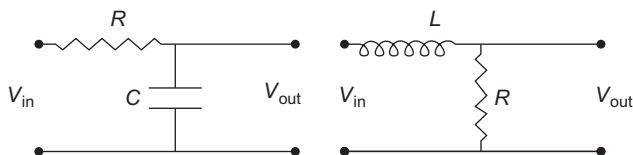


FIGURE 20.10 Circuits for Exercise 20.5.1 (left) and Exercise 20.5.2 (right).

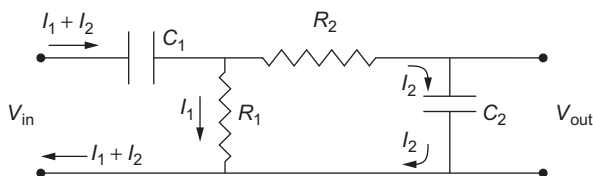


FIGURE 20.11 Circuit for Exercise 20.5.3.

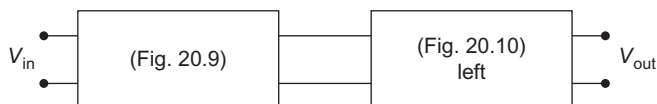


FIGURE 20.12 Representation of the circuit in Fig. 20.11 in terms of successive transfer functions.

- 20.5.3** Find the transfer function $\varphi(\omega)$ for the circuit shown in Fig. 20.11. This is a **band-pass filter**.

Hint. Assume the currents in the various parts of the circuit to have the values shown in the figure.

- 20.5.4** The circuit elements for Exercise 20.5.3 correspond to the successive transfer functions shown in Fig. 20.12. Explain why the transfer function for this exercise is only the product of the individual transfer functions in the limit $R_2 \gg R_1$.

20.6 DISCRETE FOURIER TRANSFORM

For many physicists the Fourier transform is automatically the continuous Fourier transform whose analytical properties we have been discussing in previous sections of this chapter. The use of digital computers, however, presents an opportunity to work with numerically determined Fourier transforms, which consist of values given at a discrete set of points. Integrations are therefore converted into finite summations. Transforms defined on discrete point sets have properties worth pursuing, and analysis in that area is the topic of this section.

Orthogonality on Discrete Point Sets

Throughout the earlier chapters of this book we have introduced and made use of the properties of orthogonal functions, where orthogonality has been defined as the vanishing of an integral whose integrand contains a product of the functions under study. The alternative, to be discussed here, is to define orthogonality over a discrete point set as the vanishing of a sum of products computed at the individual points. It turns out that sines, cosines, and imaginary exponentials have the remarkable property that they are also orthogonal over a series of discrete, equally spaced points on an orthogonality interval.

To analyze this situation, we take a set of N equally spaced points x_k , on the interval $(0, 2\pi)$:

$$x_k = \frac{2\pi k}{N}, \quad k = 0, 1, 2, \dots, N-1, \quad (20.115)$$

and we consider functions $\varphi_p(x)$, defined only on the points x_k and for integer p , as

$$\varphi_p(x) = e^{ipx}. \quad (20.116)$$

In line with our introductory discussion, we define the scalar products of these functions as

$$\langle \varphi_p | \varphi_q \rangle = \sum_{k=0}^{N-1} \varphi_p^*(x_k) \varphi_q(x_k). \quad (20.117)$$

Inserting [Eq. \(20.115\)](#) for the x_k , the scalar product takes the form

$$\langle \varphi_p | \varphi_q \rangle = \sum_{k=0}^{N-1} e^{2\pi i k(q-p)/N} = \sum_{k=0}^{N-1} r^k, \quad (20.118)$$

where $r = e^{2\pi i(q-p)/N}$. This is a finite geometric series; if $r = 1$ its sum has the value N ; otherwise the sum evaluates to $(1 - r^N)/(1 - r)$. But $r^N = e^{2\pi i(q-p)}$, and because p and q were restricted to integer values, we have $r^N = 1$, so the sum vanishes. To complete our understanding of the situation, we need to determine the conditions under which $r = 1$. We clearly have $r = 1$ when $q = p$. Note that we also have $r = 1$ when $q - p$ is any integer multiple of N . Thus, a formal statement relative to this scalar product is

$$\langle \varphi_p | \varphi_q \rangle = N \sum_{n=-\infty}^{\infty} \delta_{q-p, nN}. \quad (20.119)$$

Note that at most only one of the infinite sum of Kronecker deltas will be nonzero, and all will be zero unless $q - p$ is a multiple of N (one of which is $q - p = 0$).

[Equation \(20.119\)](#) is more complicated than necessary. Because the functions φ_p are defined by their values at N points, only N of them are linearly independent. In fact,

$$\varphi_{p+N}(x_k) = e^{2\pi i(p+N)k/N} = e^{2\pi i p k/N} = \varphi_p(x_k).$$

We can therefore restrict p and q in Eq. (20.119) to the range $(0, N - 1)$, and our orthogonality relation then becomes

$$\langle \varphi_p | \varphi_q \rangle = N \delta_{pq}, \quad 0 \leq p, q \leq N - 1. \quad (20.120)$$

Obviously, if function values on a discrete point set are to be used to represent a continuous function, the amount of detail that is retained in the analysis will depend on the size of the point set. We will come back to this issue in a later part of the present section.

Discrete Fourier Transform

By analogy with the definitions introduced for the conventional Fourier transform, we define the discrete transform g_p ($p = 0, \dots, N - 1$) of a function f defined only on the points x_k by the formula

$$g_p = N^{-1/2} \sum_{k=0}^{N-1} e^{2\pi i k p / N} f_k. \quad (20.121)$$

We are now writing f_k as a shorthand for $f(x_k)$, and that substitution has pretty much decoupled the problem from the original interval of definition $0 \leq x \leq 2\pi$. In essence, we are now discussing transformations between two N -member sets of function values.

The transformation inverse to Eq. (20.121) is

$$f_j = N^{-1/2} \sum_{p=0}^{N-1} e^{-2\pi i j p / N} g_p; \quad (20.122)$$

Eq. (20.122) can be verified by substituting into it the formula for g_p , yielding

$$f_j = N^{-1} \sum_{p=0}^{N-1} \sum_{k=0}^{N-1} e^{2\pi i (k-j)p / N} f_k = \sum_{k=0}^{N-1} \delta_{kj} f_k = f_j,$$

as required.

These discrete transforms have properties similar to those of their continuous cousins. For example, the transform of f_{k-j} , where j is an integer, corresponding to translation by j steps in the f array, is

$$[f_{k-j}]_p^T = N^{-1/2} \sum_{k=0}^{N-1} e^{2\pi i k p / N} f_{k-j} = e^{2\pi i j p / N} N^{-1/2} \sum_{k=0}^{N-1} e^{2\pi i (k-j)p / N} f_{k-j}.$$

Because of the periodicity of the f_k , we note that

$$N^{-1/2} \sum_{k=0}^{N-1} e^{2\pi i (k-j)p / N} f_{k-j} = N^{-1/2} \sum_{k'=-j}^{N-j-1} e^{2\pi i k' p / N} f_{k'} = N^{-1/2} \sum_{k'=0}^{N-1} e^{2\pi i k' p / N} f_{k'},$$

which is the formula for the p coefficient in the transform of f . We therefore have the translation formula

$$[f_{k-j}]_p^T = e^{2\pi i j p / N} g_p. \quad (20.123)$$

We examine next the convolution theorem, where the discrete convolution of two point sets f and g is defined as

$$[f * g]_k = N^{-1/2} \sum_{j=0}^{N-1} f_j g_{k-j}. \quad (20.124)$$

Taking the transform of this convolution, we have

$$\begin{aligned} N^{-1} \sum_{k=0}^{N-1} e^{2\pi i k p / N} \sum_{j=0}^{N-1} f_j g_{k-j} \\ = \left[N^{-1/2} \sum_{j=0}^{N-1} e^{2\pi i j p / N} f_j \right] \left[N^{-1/2} \sum_{k=0}^{N-1} e^{2\pi i (k-j) p / N} g_{k-j} \right]. \end{aligned}$$

As in the continuous case, we have split the complex exponential into two factors. We now redefine the index of the second summation from k to $l = k - j$, thereby making the two square brackets completely independent. Each can then be recognized as a transform (for the second, we need to use the fact that the g_k are periodic). The final result is

$$[f * g]_p^T = F_p G_p, \quad (20.125)$$

where F and G are the respective discrete transforms of f and g . This result is completely analogous with the convolution theorem for the continuous transform.

We close this discussion with the observation that the discrete transform and its inverse are linear transformations on coefficient arrays (vectors) of finite dimension N . Therefore, each transform operator can be represented as an $N \times N$ matrix whose rows and columns correspond to the points k or p . The fact that the transform and its inverse are complex conjugates means that the transformation matrices are unitary. Moreover, from the forms of the transform and its inverse, we see that all the elements of these matrices are proportional to complex exponentials.

Limitations

As mentioned earlier, the ability of discrete transforms to reproduce phenomena that are actually based on continuous functions will depend on the size of the point set in use. A large amount of detail on errors and limitations in the use of the discrete Fourier transform is provided by Hamming (see Additional Readings). We illustrate the potential problems in the following example.

Example 20.6.1 DISCRETE FOURIER TRANSFORM: ALIASING

Let's consider the simple case $f(x) = \cos 3x$ on the interval $0 \leq x \leq 2\pi$, which we (ill-advisedly) attempt to treat by the discrete Fourier transform method with $N = 4$. Our four points are at $x = 0, \pi/2, \pi$, and $3\pi/2$, and the four corresponding values of f_k are $(1, 0, -1, 0)$. The problem is that these same four values would be produced from $g(x) = \cos x$, so neither our discrete transform nor any information derived therefrom can properly reflect any difference in behavior between $f(x)$ and $g(x)$. If all that we are given are the four values $(1, 0, -1, 0)$, the most straightforward thing to do is take the discrete transform, yielding $(0, 1, 0, 1)$, which (from the formula for the inverse transform) corresponds to

$$\frac{1}{2}(0, 1, 0, 1) \longrightarrow \frac{e^{i\pi x/2} + e^{3i\pi x/2}}{2}.$$

If evaluated only at the chosen points, this expression is correct, but if used as an approximation over the continuous range $(0, 2\pi)$ it cannot distinguish between $\cos x$, $\cos 3x$, or any linear combination of the two with unit overall weight.

Situations in which the behavior at one wavelength or frequency is mistaken for that at another is called **aliasing**. The best way to avoid aliasing errors is to use point sets of sufficient size to accommodate the expected extent of oscillatory character in our problem. ■

Fast Fourier Transform

The fast Fourier transform (FFT) is a particular way of factoring and rearranging the terms in the sums of the discrete Fourier transform. Brought to the attention of the scientific community by Cooley and Tukey,⁴ its importance lies in the drastic reduction in the number of numerical operations required. The reduction is possible because the transformation matrix contains large numbers of duplicate entries, and the FFT procedure organizes the computation in a way permitting identical sets of coefficients to be computed only once. Because of the tremendous increase in speed achieved (and reduction in cost), the fast Fourier transform has been hailed as one of the few really significant advances in numerical analysis in the past few decades.

For N data points, a direct calculation of a discrete Fourier transform would require about N^2 multiplications. For N a power of 2, the fast Fourier transform technique of Cooley and Tukey cuts the number of multiplications required to $(N/2)\log_2 N$. If $N = 1024$ (2^{10}), the fast Fourier transform achieves a computational reduction by a factor of over 200. This is why the fast Fourier transform is called fast and why it has revolutionized

⁴J. W. Cooley and J. W. Tukey, *Math. Comput.* **19**: 297 (1965).

the digital processing of waveforms. Details on its internal operation will be found in the paper by Cooley and Tukey and in other sources.⁵

Exercises

20.6.1 Derive the trigonometric forms of discrete orthogonality corresponding to Eq. (20.120):

$$\begin{aligned}\sum_{k=0}^{N-1} \cos(2\pi pk/N) \sin(2\pi qk/N) &= 0 \\ \sum_{k=0}^{N-1} \cos(2\pi pk/N) \cos(2\pi qk/N) &= \begin{cases} 0, & p \neq q \\ N/2, & p = q \neq 0, N/2 \\ N, & p = q = 0, N/2 \end{cases} \\ \sum_{k=0}^{N-1} \sin(2\pi pk/N) \sin(2\pi qk/N) &= \begin{cases} 0, & p \neq q \\ N/2, & p = q \neq 0, N/2 \\ 0, & p = q = 0, N/2. \end{cases}\end{aligned}$$

Note. If N is odd, p and q will never have the value $N/2$.

Hint. Consider the use of trigonometric identities such as

$$\sin A \cos B = \frac{1}{2} [\sin(A+B) + \sin(A-B)].$$

20.6.2 Show in detail how to go from

$$F_p = \frac{1}{N^{1/2}} \sum_{k=0}^{N-1} f_k e^{2\pi i p k} \quad \text{to} \quad f_k = \frac{1}{N^{1/2}} \sum_{p=0}^{N-1} F_p e^{-2\pi i p k}.$$

20.6.3 The N -membered point sets f_k and F_p are discrete Fourier transforms of each other. Derive the following symmetry relations:

- (a) If f_k is real, F_p is Hermitian symmetric; that is, $F_p = F_{N-p}^*$.
- (b) If f_k is pure imaginary, then $F_p = -F_{N-p}^*$.

Note. The symmetry of part (a) is an illustration of aliasing. If F_p describes an amplitude at a frequency proportional to p , we necessarily predict an equal amplitude at the frequency proportional to $N - p$.

⁵See, for example, G. D. Bergland, A guided tour of the fast Fourier transform, *IEEE Spectrum* 6: 41 (1969). A good discussion can also be found in W. H. Press, B. P. Flannery, S. A. Teukolsky, and W. T. Vetterling, *Numerical Recipes*, 2nd ed., Cambridge: Cambridge University Press (1996), section 12.3.

20.7 LAPLACE TRANSFORMS

Definition

The Laplace transform $f(s)$ of a function $F(t)$ is defined by⁶

$$f(s) = \mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt. \quad (20.126)$$

A few comments on the existence of the integral are in order. The infinite integral of $F(t)$,

$$\int_0^{\infty} F(t) dt,$$

need not exist. For instance, $F(t)$ may diverge exponentially for large t . However, if there are some constants s_0 , M , and $t_0 \geq 0$ such that for all $t > t_0$

$$|e^{-s_0 t} F(t)| \leq M, \quad (20.127)$$

the Laplace transform will exist for $s > s_0$; $F(t)$ is then said to be of **exponential order**. As a counterexample, $F(t) = e^{t^2}$ does not satisfy the condition given by Eq. (20.127) and is **not** of exponential order. Thus, $\mathcal{L}\{e^{t^2}\}$ does **not** exist.

The Laplace transform may also fail to exist because of a sufficiently strong singularity in the function $F(t)$ as $t \rightarrow 0$. For example,

$$\int_0^{\infty} e^{-st} t^n dt$$

diverges at the origin for $n \leq -1$. The Laplace transform $\mathcal{L}\{t^n\}$ does not exist for $n \leq -1$.

Since, for two functions $F(t)$ and $G(t)$ for which the integrals exist,

$$\mathcal{L}\{aF(t) + bG(t)\} = a\mathcal{L}\{F(t)\} + b\mathcal{L}\{G(t)\}, \quad (20.128)$$

the operation denoted by $\mathcal{L}$ is **linear**.

Elementary Functions

To introduce the Laplace transform, let us apply the operation to some of the elementary functions. In all cases we assume that $F(t) = 0$ for $t < 0$. If

$$F(t) = 1, \quad t > 0,$$

⁶This is sometimes called a **one-sided Laplace transform**; the integral from $-\infty$ to $+\infty$ is referred to as a **two-sided Laplace transform**. Some authors introduce an additional factor of s . This extra s appears to have little advantage and continually gets in the way; for further comments, see section 14.13 in the text by Jeffreys and Jeffreys (Additional Readings). Generally, we take s to be real and positive. It is possible to let s become complex, provided $\Re(s) > 0$.

then

$$\mathcal{L}\{1\} = \int_0^{\infty} e^{-st} dt = \frac{1}{s}, \quad \text{for } s > 0. \quad (20.129)$$

Next, let

$$F(t) = e^{kt}, \quad t > 0.$$

The Laplace transform becomes

$$\mathcal{L}\{e^{kt}\} = \int_0^{\infty} e^{-st} e^{kt} dt = \frac{1}{s-k}, \quad \text{for } s > k. \quad (20.130)$$

Using this relation, we obtain the Laplace transform of certain other functions. Since

$$\cosh kt = \frac{1}{2}(e^{kt} + e^{-kt}), \quad \sinh kt = \frac{1}{2}(e^{kt} - e^{-kt}), \quad (20.131)$$

we have

$$\mathcal{L}\{\cosh kt\} = \frac{1}{2} \left(\frac{1}{s-k} + \frac{1}{s+k} \right) = \frac{s}{s^2 - k^2}, \quad (20.132)$$

$$\mathcal{L}\{\sinh kt\} = \frac{1}{2} \left(\frac{1}{s-k} - \frac{1}{s+k} \right) = \frac{k}{s^2 - k^2}, \quad (20.133)$$

both valid for $s > k$.

From the relations

$$\cos kt = \cosh ikt, \quad \sin kt = -i \sinh ikt,$$

it is evident that we can obtain transforms of the sine and cosine if k is replaced by ik in Eqs. (20.132) and (20.133):

$$\mathcal{L}\{\cos kt\} = \frac{s}{s^2 + k^2}, \quad (20.134)$$

$$\mathcal{L}\{\sin kt\} = \frac{k}{s^2 + k^2}, \quad (20.135)$$

both valid for $s > 0$. Another derivation of this last transform is given in [Example 20.8.1](#). It is a curious fact that $\lim_{s \rightarrow 0} \mathcal{L}\{\sin kt\} = 1/k$ despite the fact that $\int_0^{\infty} \sin kt dt$ does not exist.

Finally, for $F(t) = t^n$, we have

$$\mathcal{L}\{t^n\} = \int_0^{\infty} e^{-st} t^n dt,$$

which is just a gamma function. Hence

$$\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}}, \quad s > 0, \quad n > -1. \quad (20.136)$$

Note that in all these transforms we have the variable s in the denominator, so that it occurs as a negative power. From the definition of the transform, Eq. (20.126) and the existence condition, Eq. (20.127), it is clear that if $f(s)$ is a Laplace transform, then $\lim_{s \rightarrow \infty} f(s) = 0$. The significance of this point is that if $f(s)$ behaves asymptotically for large s as a positive power of s , then no inverse transform can exist.

Heaviside Step Function

In Exercise 1.11.9 we encountered the Heaviside step function $u(t)$. Because of its utility in describing discontinuous signal pulses, its Laplace transform occurs frequently. We therefore remind the reader of the definition

$$u(t - k) = \begin{cases} 0, & t < k, \\ 1, & t > k. \end{cases} \quad (20.137)$$

Taking the transform, we have

$$\mathcal{L}\{u(t - k)\} = \int_k^{\infty} e^{-st} dt = \frac{1}{s} e^{-ks}. \quad (20.138)$$

Example 20.7.1 TRANSFORM OF SQUARE PULSE

Let's compute the transform of a square pulse $F(t)$ of height A that is on from $t = 0$ to $t = t_0$; see Fig. 20.13. Using the Heaviside step function, the pulse can be represented as

$$F(t) = A[u(t) - u(t - t_0)].$$

Its transform is therefore

$$\mathcal{L}\{F(t)\} = \frac{1}{s} (1 - e^{-t_0 s}).$$

■

Dirac Delta Function

For use with differential equations one further transform is helpful, namely that of the Dirac delta function. From the properties of the delta function, we have

$$\mathcal{L}\{\delta(t - t_0)\} = \int_0^{\infty} e^{-st} \delta(t - t_0) dt = e^{-st_0}, \quad \text{for } t_0 > 0. \quad (20.139)$$

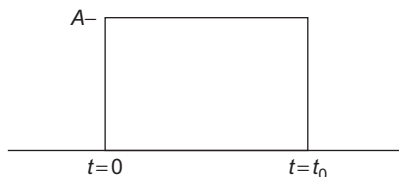


FIGURE 20.13 Square pulse.

For $t_0 = 0$ we must be a bit more careful, as the sequences we have used for defining the delta function involve contributions symmetrically distributed about t_0 , and the integration defining the Laplace transform is restricted to $t \geq 0$. Consistent results when using Laplace transforms, however, are obtained if we consider delta sequences that are entirely within the range $t \geq t_0$, which is equivalent to

$$\mathcal{L}\{\delta(t)\} = 1. \quad (20.140)$$

This delta function is frequently called the **impulse** function because it is so useful in describing impulsive forces, that is, forces lasting only a short time.

Inverse Transform

As we have already seen in our discussion of the Fourier transform, the taking of an integral transform will ordinarily have little value unless we can carry out the inverse transform. That is, with

$$\mathcal{L}\{F(t)\} = f(s),$$

then it is desirable to be able to compute

$$\mathcal{L}^{-1}\{f(s)\} = F(t). \quad (20.141)$$

However, this inverse transform is not entirely unique. Two functions $F_1(t)$ and $F_2(t)$ can have the same transform, $f(s)$, if their difference, $N(t) = F_1(t) - F_2(t)$, is a **null function**, meaning that for all $t_0 > 0$ it satisfies

$$\int_0^{t_0} N(t) dt = 0.$$

This result is known as **Lerch's theorem**, and is not quite equivalent to $F_1 = F_2$, because it permits F_1 and F_2 to differ at isolated points. However, in most problems studied by physicists or engineers this ambiguity is not important and we will not consider it further.

The inverse transform can be determined in various ways.

1. A table of transforms can be built up and used to identify inverse transformations, exactly as a table of logarithms can be used to look up antilogarithms. The preceding transforms constitute the beginnings of such a table. More complete sets of Laplace transforms are in several of the Additional Readings, and a relatively short table of transforms appears in the present text as [Table 20.1](#). Many functional forms not in [Table 20.1](#) can be reduced to tabular entries using a partial fraction expansion or other properties of the Laplace transform presented later in this chapter. Of particular value in this regard are the translation and derivative formulas. There is some justification for suspecting that these tables are probably of more value in solving textbook exercises than in solving real-world problems.
2. A general technique for $\mathcal{L}^{-1}$ will be developed in [Section 20.10](#) by using the calculus of residues.
3. Transforms and their inverses can be represented numerically. See the work by Krylov and Skoblya in Additional Readings.

Table 20.1 Laplace Transforms^a

	$f(s)$	$F(t)$	Limitation	Equation
1.	1	$\delta(t)$	Singularity at +0	(20.140)
2.	$\frac{1}{s}$	1	$s > 0$	(20.129)
3.	$\frac{\Gamma(n+1)}{s^{n+1}}$	t^n	$s > 0, n > -1$	(20.136)
4.	$\frac{1}{s-k}$	e^{kt}	$s > k$	(20.130)
5.	$\frac{1}{(s-k)^2}$	te^{kt}	$s > k$	(20.176)
6.	$\frac{s}{s^2-k^2}$	$\cosh kt$	$s > k$	(20.132)
7.	$\frac{k}{s^2-k^2}$	$\sinh kt$	$s > k$	(20.133)
8.	$\frac{s}{s^2+k^2}$	$\cos kt$	$s > 0$	(20.134)
9.	$\frac{k}{s^2+k^2}$	$\sin kt$	$s > 0$	(20.135)
10.	$\frac{s-a}{(s-a)^2+k^2}$	$e^{at} \cos kt$	$s > a$	(20.159)
11.	$\frac{k}{(s-a)^2+k^2}$	$e^{at} \sin kt$	$s > a$	(20.158)
12.	$\frac{s^2-k^2}{(s^2+k^2)^2}$	$t \cos kt$	$s > 0$	(20.177)
13.	$\frac{2ks}{(s^2+k^2)^2}$	$t \sin kt$	$s > 0$	(20.178)
14.	$(s^2+a^2)^{-1/2}$	$J_0(at)$	$s > 0$	(20.182)
15.	$(s^2-a^2)^{-1/2}$	$I_0(at)$	$s > a$	Exercise 20.8.13
16.	$\frac{1}{a} \cot^{-1} \left(\frac{s}{a} \right)$	$j_0(at)$	$s > 0$	Exercise 20.8.14
17.	$\left. \begin{array}{l} \frac{1}{2a} \ln \frac{s+a}{s-a} \\ \frac{1}{a} \coth^{-1} \left(\frac{s}{a} \right) \end{array} \right\}$	$i_0(at)$	$s > a$	Exercise 20.8.14
18.	$\frac{(s-a)^n}{s^{n+1}}$	$L_n(at)$	$s > 0$	Exercise 20.8.16
19.	$\frac{1}{s} \ln(s+1)$	$E_1(x)$	$s > 0$	Exercise 20.8.17
20.	$\frac{\ln s}{s}$	$-\ln t - \gamma$	$s > 0$	Exercise 20.10.9

^a γ is the Euler-Mascheroni constant.

Example 20.7.2 PARTIAL FRACTION EXPANSION

The function $f(s) = k^2/s(s^2 + k^2)$ does not appear as a transform listed in Table 20.1, but we may obtain it from the tabulated transforms by observing that it has the partial fraction expansion

$$f(s) = \frac{k^2}{s(s^2 + k^2)} = \frac{1}{s} - \frac{s}{s^2 + k^2}.$$

The partial fraction technique was discussed in Section 1.5, and the present example was the subject of Example 1.5.3.

Each of the two partial fractions corresponds to an entry in Table 20.1, and we can therefore take the inverse transform of $f(s)$ term by term:

$$\mathcal{L}^{-1}\{f(s)\} = 1 - \cos kt.$$

Remember that the range of the inverse transform is restricted to $t \geq 0$. ■

Example 20.7.3 A STEP FUNCTION

This example shows how Laplace transforms can be used to evaluate a definite integral. Consider

$$F(t) = \int_0^{\infty} \frac{\sin tx}{x} dx. \quad (20.142)$$

Suppose we take the Laplace transform of this definite (and improper) integral, naming it $f(s)$:

$$f(s) = \mathcal{L} \left\{ \int_0^{\infty} \frac{\sin tx}{x} dx \right\} = \int_0^{\infty} e^{-st} \int_0^{\infty} \frac{\sin tx}{x} dx dt.$$

Now, interchanging the order of integration (which is justified),⁷ we get

$$f(s) = \int_0^{\infty} \frac{1}{x} \left[\int_0^{\infty} e^{-st} \sin tx dt \right] dx = \int_0^{\infty} \frac{dx}{s^2 + x^2}, \quad (20.143)$$

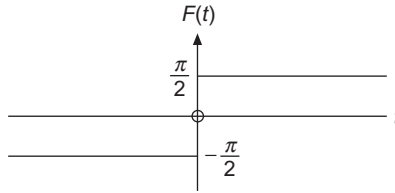
since the factor in square brackets is just the Laplace transform of $\sin tx$. The integral on the right-hand side is elementary, with evaluation

$$f(s) = \int_0^{\infty} \frac{dx}{s^2 + x^2} = \frac{1}{s} \tan^{-1} \left(\frac{x}{s} \right) \Big|_0^{\infty} = \frac{\pi}{2s}. \quad (20.144)$$

Using entry #2 in Table 20.1, we carry out the inverse transformation to obtain

$$F(t) = \frac{\pi}{2}, \quad t > 0, \quad (20.145)$$

⁷See Chapter 1 in Jeffreys and Jeffreys (Additional Readings) for a discussion of uniform convergence of integrals.

FIGURE 20.14 $F(t) = \int_0^\infty \frac{\sin tx}{x} dx$, a step function.

in agreement with an evaluation by the calculus of residues, Eq. (11.107). It has been assumed that $t > 0$ in $F(t)$. For $F(-t)$ we need note only that $\sin(-tx) = -\sin tx$, giving $F(-t) = -F(t)$. Finally, if $t = 0$, $F(0)$ is clearly zero. Therefore,

$$\int_0^\infty \frac{\sin tx}{x} dx = \frac{\pi}{2} [2u(t) - 1] = \begin{cases} \frac{\pi}{2}, & t > 0 \\ 0, & t = 0 \\ -\frac{\pi}{2}, & t < 0. \end{cases} \quad (20.146)$$

Here $u(t)$ is the Heaviside unit step function, Eq. (20.137). Thus, $\int_0^\infty (\sin tx/x) dx$, taken as a function of t , describes a step function (Fig. 20.14), with a step of height π at $t = 0$. ■

The technique in the preceding example was to (1) introduce a second integration, namely the Laplace transform, (2) reverse the order of integration and integrate once, and (3) take the inverse Laplace transform. This is a technique that will apply to many problems.

Exercises

20.7.1 Prove that

$$\lim_{s \rightarrow \infty} sf(s) = \lim_{t \rightarrow +0} F(t).$$

Hint. Assume that $F(t)$ can be expressed as $F(t) = \sum_{n=0}^\infty a_n t^n$.

20.7.2 Show that

$$\frac{1}{\pi} \lim_{s \rightarrow 0} \mathcal{L}\{\cos xt\} = \delta(x).$$

20.7.3 Verify that

$$\mathcal{L}\left\{\frac{\cos at - \cos bt}{b^2 - a^2}\right\} = \frac{s}{(s^2 + a^2)(s^2 + b^2)}, \quad a^2 \neq b^2.$$

20.7.4 Using partial fraction expansions, show that

$$(a) \quad \mathcal{L}^{-1} \left\{ \frac{1}{(s+a)(s+b)} \right\} = \frac{e^{-at} - e^{-bt}}{b-a}, \quad a \neq b.$$

$$(b) \quad \mathcal{L}^{-1} \left\{ \frac{s}{(s+a)(s+b)} \right\} = \frac{ae^{-at} - be^{-bt}}{a-b}, \quad a \neq b.$$

20.7.5 Using partial fraction expansions, show that for $a^2 \neq b^2$,

$$(a) \quad \mathcal{L}^{-1} \left\{ \frac{1}{(s^2+a^2)(s^2+b^2)} \right\} = -\frac{1}{a^2-b^2} \left\{ \frac{\sin at}{a} - \frac{\sin bt}{b} \right\}.$$

$$(b) \quad \mathcal{L}^{-1} \left\{ \frac{s^2}{(s^2+a^2)(s^2+b^2)} \right\} = \frac{1}{a^2-b^2} \{a \sin at - b \sin bt\}.$$

20.7.6 Show that

$$(a) \quad \int_0^\infty \frac{\cos s}{s^\nu} ds = \frac{\pi}{2(\nu-1)! \cos(\nu\pi/2)}, \quad 0 < \nu < 1.$$

$$(b) \quad \int_0^\infty \frac{\sin s}{s^\nu} ds = \frac{\pi}{2(\nu-1)! \sin(\nu\pi/2)}, \quad 0 < \nu < 2.$$

Why is ν restricted to $(0, 1)$ for (a), to $(0, 2)$ for (b)? These integrals may be interpreted as Fourier transforms of $s^{-\nu}$ and as Mellin transforms of $\sin s$ and $\cos s$.

Hint. Replace $s^{-\nu}$ by a Laplace transform integral: $\mathcal{L}\{t^{\nu-1}\} / \Gamma(\nu)$. Then integrate with respect to s . The resulting integral can be treated as a beta function (Section 13.3).

20.7.7 A function $F(t)$ can be expanded in a Maclaurin series,

$$F(t) = \sum_{n=0}^{\infty} a_n t^n.$$

Then

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st} \sum_{n=0}^{\infty} a_n t^n dt = \sum_{n=0}^{\infty} a_n \int_0^\infty e^{-st} t^n dt.$$

Show that $f(s)$, the Laplace transform of $F(t)$, contains no powers of s greater than s^{-1} . Check your result by calculating $\mathcal{L}\{\delta(t)\}$, and comment on this fiasco.

20.7.8 Show that the Laplace transform of the confluent hypergeometric function $M(a, c; x)$ is

$$\mathcal{L}\{M(a, c; x)\} = \frac{1}{s} {}_2F_1\left(a, 1; c; \frac{1}{s}\right).$$

20.8 PROPERTIES OF LAPLACE TRANSFORMS

Transforms of Derivatives

Perhaps the main application of Laplace transforms is in converting differential equations into simpler forms that may be solved more easily. It will be seen, for instance, that coupled differential equations with constant coefficients transform to simultaneous linear algebraic equations. For the study of differential equations we need formulas for the Laplace transforms of the derivatives of a function.

Let us transform the first derivative of $F(t)$:

$$\mathcal{L}\{F'(t)\} = \int_0^{\infty} e^{-st} \frac{dF(t)}{dt} dt.$$

Integrating by parts, we obtain

$$\begin{aligned} \mathcal{L}\{F'(t)\} &= e^{-st} F(t) \Big|_0^{\infty} + s \int_0^{\infty} e^{-st} F(t) dt \\ &= s\mathcal{L}\{F(t)\} - F(0). \end{aligned} \tag{20.147}$$

Strictly speaking, $F(0) = F(+0)$,⁸ and dF/dt is required to be at least piecewise continuous for $0 \leq t < \infty$. Naturally, both $F(t)$ and its derivative must be such that the integrals do not diverge. An extension to higher derivatives gives

$$\mathcal{L}\{F^{(2)}(t)\} = s^2 \mathcal{L}\{F(t)\} - sF(+0) - F'(+0), \tag{20.148}$$

$$\mathcal{L}\{F^{(n)}(t)\} = s^n \mathcal{L}\{F(t)\} - s^{n-1} F(+0) - \dots - F^{(n-1)}(+0). \tag{20.149}$$

The Laplace transform, like the Fourier transform, replaces differentiation with multiplication. In the following examples ODEs become algebraic equations. Here is the power and the utility of the Laplace transform. But see [Example 20.8.7](#) for what may happen if the coefficients are not constant.

Note how the initial conditions, $F(+0)$, $F'(+0)$, and so on, are incorporated into the transform. This situation is different than for the Fourier transform, and arises from the finite lower limit ($t = 0$) of the integral defining the transform. This property makes the Laplace transform more powerful for obtaining solutions to differential equations subject to initial conditions.

⁸This notation means that zero is approached from the positive side.

Example 20.8.1 USE OF DERIVATIVE FORMULA

Here is an example showing how the derivative formula has uses even in contexts not involving the solution to a differential equation. Starting from the identity

$$-k^2 \sin kt = \frac{d^2}{dt^2} \sin kt, \quad (20.150)$$

we apply on both sides of the equation the Laplace transform operation, reaching

$$\begin{aligned} -k^2 \mathcal{L}\{\sin kt\} &= \mathcal{L}\left\{\frac{d^2}{dt^2} \sin kt\right\} \\ &= s^2 \mathcal{L}\{\sin kt\} - s \sin(0) - \left.\frac{d}{dt} \sin kt\right|_{t=0}. \end{aligned}$$

Since $\sin(0) = 0$ and $d/dt \sin kt|_{t=0} = k$, the above equation has solution

$$\mathcal{L}\{\sin kt\} = \frac{k}{s^2 + k^2}.$$

This result confirms [Eq. \(20.135\)](#). ■

Examples involving the solutions of differential equations follow.

Example 20.8.2 SIMPLE HARMONIC OSCILLATOR

As a physical example, consider a mass m oscillating under the influence of an ideal spring, spring constant k . As usual, friction is neglected. Then Newton's second law becomes

$$m \frac{d^2 X(t)}{dt^2} + kX(t) = 0. \quad (20.151)$$

We take as initial conditions

$$X(0) = X_0, \quad X'(0) = 0.$$

Applying the Laplace transform, we obtain

$$m \mathcal{L}\left\{\frac{d^2 X}{dt^2}\right\} + k \mathcal{L}\{X(t)\} = 0. \quad (20.152)$$

Letting $x(s)$ denote the presently unknown transform $\mathcal{L}\{X(t)\}$ and using [Eq. \(20.148\)](#), we convert [Eq. \(20.152\)](#) to the form

$$ms^2 x(s) - msX_0 + kx(s) = 0,$$

which has solution

$$x(s) = X_0 \frac{s}{s^2 + \omega_0^2}, \quad \text{with } \omega_0^2 \equiv \frac{k}{m}.$$

From [Table 20.1](#) this is seen to be the transform of $\cos \omega_0 t$, which gives the expected result:

$$X(t) = X_0 \cos \omega_0 t. \quad (20.153)$$
■

Example 20.8.3 EARTH'S NUTATION

A somewhat more involved example is the nutation of the Earth's poles (force-free precession). We treat the Earth as a rigid (oblate) spheroid, with z -axis through its direction of symmetry. We assume the spheroid to have moments of inertia I_z and $I_x = I_y$ and to be rotating about its x , y , and z axes at the respective angular velocities $X(t) = \omega_x(t)$, $Y(t) = \omega_y(t)$, $\omega_z = \text{constant}$. The Euler equations of motion for X and Y reduce to

$$\frac{dX}{dt} = -aY, \quad \frac{dY}{dt} = +aX, \quad (20.154)$$

where $a \equiv [(I_z - I_x)/I_z]\omega_z$. For the Earth, the initial values of X and Y are not both zero, so the axis of rotation is not aligned with the symmetry axis (see Fig. 20.15), and because of this lack of alignment, the axis of rotation precesses about the axis of symmetry. For the Earth, the deviation between the rotation and symmetry axes is small, only about 15 meters (measured at the Earth's surface at the poles).

Our first step in solving these coupled ODEs is to take their Laplace transforms, obtaining

$$sx(s) - X(0) = -ay(s), \quad sy(s) - Y(0) = ax(s).$$

Combining to eliminate $y(s)$, we have

$$s^2x(s) - sX(0) + aY(0) = -a^2x(s),$$

or

$$x(s) = X(0)\frac{s}{s^2 + a^2} - Y(0)\frac{a}{s^2 + a^2}. \quad (20.155)$$

Recognizing these functions of s as transforms listed in Table 20.1,

$$X(t) = X(0)\cos at - Y(0)\sin at.$$

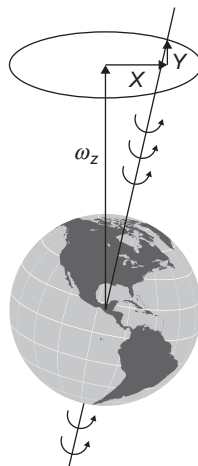


FIGURE 20.15 Earth's rotation axis and its components.

Similarly,

$$Y(t) = X(0)\sin at + Y(0)\cos at.$$

This is seen to be a rotation of the vector (X, Y) counterclockwise (for $a > 0$) about the z -axis with angle $\theta = at$ and angular velocity a .

A direct interpretation may be found by choosing the x and y axes so that $Y(0) = 0$. Then

$$X(t) = X(0)\cos at, \quad Y(t) = X(0)\sin at,$$

which are the parametric equations for rotation of (X, Y) in a circular orbit of radius $X(0)$, with angular velocity a in the counterclockwise sense.

For the Earth, a as defined here corresponds to a period $(2\pi/a)$ of some 300 days. Actually, because of departures from the idealized rigid body assumed in setting up Euler's equations, the period is about 427 days.⁹

These same equations arise in electromagnetic theory. If in Eq. (20.154) we set

$$X(t) = L_x, \quad Y(t) = L_y,$$

where L_x and L_y are the x - and y -components of the angular momentum $\mathbf{L}$ of a charged particle moving in a uniform magnetic field $B_z \mathbf{e}_z$, and then assign a the value $a = -g_L B_z$, where g_L is the **gyromagnetic ratio** of the particle, then Eq. (20.148) determines its Larmor precession in the magnetic field. ■

Example 20.8.4 IMPULSIVE FORCE

For an impulsive force acting on a particle of mass m , Newton's second law takes the form

$$m \frac{d^2 X}{dt^2} = P \delta(t),$$

where P is a constant. Transforming, we obtain

$$ms^2 x(s) - msX(0) - mX'(0) = P.$$

For a particle starting from rest, $X'(0) = 0$. We shall also take $X(0) = 0$. Then

$$x(s) = \frac{P}{ms^2},$$

and, taking the inverse transform,

$$X(t) = \frac{P}{m} t,$$

$$\frac{dX(t)}{dt} = \frac{P}{m}, \quad \text{a constant.}$$

The effect of the impulse $P \delta(t)$ is to transfer (instantaneously) P units of linear momentum to the particle.

⁹D. Menzel, ed., *Fundamental Formulas of Physics*, Englewood Cliffs, NJ: Prentice-Hall (1955), reprinted, Dover (1960), p. 695.

A similar analysis applies to the ballistic galvanometer. The torque on the galvanometer is given initially by $k\iota$, in which ι is a pulse of current and k is a proportionality constant. Since ι is of short duration, we set

$$k\iota = kq \delta(t),$$

where q is the total charge carried by the current ι . Then, with I the moment of inertia,

$$I \frac{d^2\theta}{dt^2} = kq \delta(t),$$

and transforming, as before, we find that the effect of the current pulse is a transfer of kq units of **angular** momentum to the galvanometer. ■

Change of Scale

If we replace t by at in the defining formula for the Laplace transform, we readily obtain

$$\begin{aligned} \mathcal{L}\{F(at)\} &= \int_0^{\infty} e^{-st} F(at) dt = \frac{1}{a} \int_0^{\infty} e^{-(s/a)(at)} F(at) d(at) \\ &= \frac{1}{a} f\left(\frac{s}{a}\right). \end{aligned} \quad (20.156)$$

Substitution

If we replace the parameter s by $s - a$ in the definition of the Laplace transform, Eq. (20.126), we have

$$\begin{aligned} f(s - a) &= \int_0^{\infty} e^{-(s-a)t} F(t) dt = \int_0^{\infty} e^{-st} e^{at} F(t) dt \\ &= \mathcal{L}\{e^{at} F(t)\}. \end{aligned} \quad (20.157)$$

Hence the replacement of s with $s - a$ corresponds to multiplying $F(t)$ by e^{at} , and conversely. This result can be used to check some entries in our table of transforms. From Eq. (20.157) we find immediately that

$$\mathcal{L}\{e^{at} \sin kt\} = \frac{k}{(s - a)^2 + k^2}, \quad (s > a), \quad (20.158)$$

and

$$\mathcal{L}\{e^{at} \cos kt\} = \frac{s - a}{(s - a)^2 + k^2}, \quad s > a. \quad (20.159)$$

These are entries 10 and 11 of Table 20.1.

Example 20.8.5 DAMPED OSCILLATOR

Equations (20.158) and (20.159) are useful when we consider an oscillating mass with damping proportional to the velocity. Equation (20.151), with such damping added, becomes

$$mX''(t) + bX'(t) + kX(t) = 0, \quad (20.160)$$

in which b is a proportionality constant. Let us assume that the particle starts from rest at $X(0) = X_0$, so $X'(0) = 0$. The transformed equation is

$$m[s^2x(s) - sX_0] + b[sx(s) - X_0] + kx(s) = 0,$$

with solution

$$x(s) = X_0 \frac{ms + b}{ms^2 + bs + k}.$$

This transform does not appear in our table, but may be handled by completing the square of the denominator:

$$s^2 + \frac{b}{m}s + \frac{k}{m} = \left(s + \frac{b}{2m}\right)^2 + \left(\frac{k}{m} - \frac{b^2}{4m^2}\right).$$

Considering further only the case that the damping is small enough that $b^2 < 4km$, then the last term is positive and will be denoted by ω_1^2 . We then rearrange $x(s)$ to the form

$$\begin{aligned} x(s) &= X_0 \frac{s + b/m}{(s + b/2m)^2 + \omega_1^2} \\ &= X_0 \frac{s + b/2m}{(s + b/2m)^2 + \omega_1^2} + X_0 \frac{\omega_1(b/2m\omega_1)}{(s + b/2m)^2 + \omega_1^2}. \end{aligned}$$

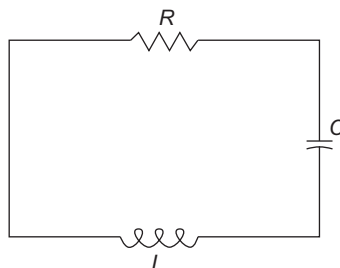
These are the same transforms we encountered in Eqs. (20.158) and (20.159), so we may take the inverse transform of our formula for $x(s)$, reaching

$$\begin{aligned} X(t) &= X_0 e^{-(b/2m)t} \left(\cos \omega_1 t + \frac{b}{2m\omega_1} \sin \omega_1 t \right) \\ &= X_0 \frac{\omega_0}{\omega_1} e^{-(b/2m)t} \cos(\omega_1 t - \varphi). \end{aligned} \quad (20.161)$$

Here we have made the substitutions

$$\tan \varphi = \frac{b}{2m\omega_1}, \quad \omega_0^2 = \frac{k}{m}.$$

Of course, as $b \rightarrow 0$, this solution goes over to the undamped solution, given in Example 20.8.2. ■

FIGURE 20.16 RLC circuit.

RLC Analog

It is worth noting the similarity between the damped simple harmonic oscillation of a mass (Example 20.8.5) and an RLC circuit (resistance, inductance, and capacitance). See Fig. 20.16. At any instant, the sum of the potential differences around the loop must be zero (Kirchhoff's law, conservation of energy). This gives

$$L \frac{dI}{dt} + RI + \frac{1}{C} \int I dt = 0. \quad (20.162)$$

Differentiating Eq. (20.162) with respect to time (to eliminate the integral), we have

$$L \frac{d^2 I}{dt^2} + R \frac{dI}{dt} + \frac{1}{C} I = 0. \quad (20.163)$$

If we replace $I(t)$ with $X(t)$, L with m , R with b , and C^{-1} with k , then Eq. (20.163) is identical with the mechanical problem. It is but one example of the unification of diverse branches of physics by mathematics. A more complete discussion will be found in a book by Olson.¹⁰

Translation

This time let $f(s)$ be multiplied by e^{-bs} , with $b > 0$:

$$\begin{aligned} e^{-bs} f(s) &= e^{-bs} \int_0^{\infty} e^{-st} F(t) dt \\ &= \int_0^{\infty} e^{-s(t+b)} F(t) dt. \end{aligned} \quad (20.164)$$

¹⁰H. F. Olson, *Dynamical Analogies*, New York: Van Nostrand (1943).

Now let $t + b = \tau$. Equation (20.164) becomes

$$e^{-bs} f(s) = \int_b^{\infty} e^{-s\tau} F(\tau - b) d\tau. \quad (20.165)$$

Since $F(t)$ is assumed to be equal to zero for $t < 0$, so that $F(\tau - b) = 0$ for $0 \leq \tau < b$, we can change the lower limit in Eq. (20.165) to zero without changing the value of the integral. Then renaming τ as our standard Laplace transform variable t , we have

$$e^{-bs} f(s) = \mathcal{L}\{F(t - b)\}. \quad (20.166)$$

If instead of relying on the assumption that $F(t) = 0$ for negative t we insert a Heaviside unit step function $u(\tau - b)$ to restrict the contributions from F to positive arguments, Eq. (20.165) takes the form

$$e^{-bs} f(s) = \int_0^{\infty} e^{-s\tau} F(\tau - b) u(\tau - b) d\tau.$$

For this reason the translation formula, Eq. (20.166), is often called the **Heaviside shifting theorem**.

Example 20.8.6 ELECTROMAGNETIC WAVES

The electromagnetic wave equation with $E = E_y$ or E_z , a transverse wave propagating along the x -axis, is

$$\frac{\partial^2 E(x, t)}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 E(x, t)}{\partial t^2} = 0. \quad (20.167)$$

We want to solve this PDE for the situation that a source at $x = 0$ generates a time-dependent signal $E(0, t)$ starting at time $t = 0$ and propagating only toward positive x , with initial conditions that for $x > 0$,

$$E(x, 0) = 0, \quad \left. \frac{\partial E(x, t)}{\partial t} \right|_{t=0} = 0.$$

Transforming Eq. (20.167) with respect to t , we get

$$\frac{\partial^2}{\partial x^2} \mathcal{L}\{E(x, t)\} - \frac{s^2}{v^2} \mathcal{L}\{E(x, t)\} + \frac{s}{v^2} E(x, 0) + \frac{1}{v^2} \left. \frac{\partial E(x, t)}{\partial t} \right|_{t=0} = 0,$$

which due to the initial conditions simplifies to

$$\frac{\partial^2}{\partial x^2} \mathcal{L}\{E(x, t)\} = \frac{s^2}{v^2} \mathcal{L}\{E(x, t)\}. \quad (20.168)$$

The general solution of Eq. (20.168) (which is an **ODE** in x) is

$$\mathcal{L}\{E(x, t)\} = f_1(s)e^{-(s/v)x} + f_2(s)e^{+(s/v)x}. \quad (20.169)$$

To understand more fully this result consider first the case $f_2(s) = 0$. Then Eq. (20.169) becomes

$$\mathcal{L}\{E(x, t)\} = e^{-(x/v)s} f_1(s), \quad (20.170)$$

which we recognize as of the same form as Eq. (20.166), meaning that

$$E(x, t) = F\left(t - \frac{x}{v}\right),$$

where F is the function whose Laplace transform is f_1 , namely $E(0, t)$.¹¹ Since F is assumed to vanish when its argument is negative, this formula can be written in the more explicit form

$$E(x, t) = \begin{cases} F\left(t - \frac{x}{v}\right) = E\left(0, t - \frac{x}{v}\right), & t \geq \frac{x}{v}, \\ 0, & t < \frac{x}{v}. \end{cases} \quad (20.171)$$

This solution represents a wave (or pulse) moving in the positive x -direction with velocity v . Note that for $x > vt$ the region remains undisturbed; the pulse has not had time to get there. If we had decided to take the solution of Eq. (20.169) with $f_1(s) = 0$, we would have obtained

$$E(x, t) = \begin{cases} F\left(t + \frac{x}{v}\right) = E\left(0, t + \frac{x}{v}\right), & t \geq -\frac{x}{v}, \\ 0, & t < -\frac{x}{v}, \end{cases} \quad (20.172)$$

which we must reject because (for propagation toward positive x) it violates causality.

Our solution to this problem, Eq. (20.171), can be verified by differentiation and substitution into the original PDE, Eq. (20.167). ■

Derivative of a Transform

When $F(t)$, which is at least piecewise continuous, and s are chosen so that $e^{-st}F(t)$ converges exponentially for large s , the integral

$$\int_0^{\infty} e^{-st} F(t) dt$$

¹¹Consider Eq. (20.170) with x set to zero.

is uniformly convergent and may be differentiated (under the integral sign) with respect to s . Then

$$f'(s) = \int_0^{\infty} (-t)e^{-st} F(t) dt = \mathcal{L}\{-tF(t)\}. \quad (20.173)$$

Continuing this process, we obtain

$$f^{(n)}(s) = \mathcal{L}\{(-t)^n F(t)\}. \quad (20.174)$$

All the integrals so obtained will be uniformly convergent because of the decreasing exponential behavior of $e^{-st} F(t)$.

This technique may be applied to generate more transforms. For example,

$$\mathcal{L}\{e^{kt}\} = \int_0^{\infty} e^{-st} e^{kt} dt = \frac{1}{s-k}, \quad s > k. \quad (20.175)$$

Differentiating with respect to s (or with respect to k), we obtain

$$\mathcal{L}\{te^{kt}\} = \frac{1}{(s-k)^2}, \quad s > k. \quad (20.176)$$

If we replace k by ik and separate Eq. (20.176) into its real and imaginary parts, we get

$$\mathcal{L}\{t \cos kt\} = \frac{s^2 - k^2}{(s^2 + k^2)^2}, \quad (20.177)$$

$$\mathcal{L}\{t \sin kt\} = \frac{2ks}{(s^2 + k^2)^2}. \quad (20.178)$$

These expressions are valid for $s > 0$.

Example 20.8.7 BESSEL'S EQUATION

An interesting application of a differentiated Laplace transform appears in the solution of Bessel's equation with $n = 0$. From Chapter 14 we have

$$x^2 y''(x) + xy'(x) + x^2 y(x) = 0.$$

This ODE cannot be solved by the method illustrated in Example 20.8.2 because the derivatives are multiplied by functions of the independent variable x . However, an alternate approach depending on Eq. (20.174) is available. Dividing by x and substituting $t = x$ and $F(t) = y(x)$ to agree with the present notation, we see that the Bessel equation becomes

$$tF''(t) + F'(t) + tF(t) = 0. \quad (20.179)$$

We need a regular solution, and it appears possible for $F(0)$ to be nonzero, so we scale the solution by setting $F(0) = 1$. Then, setting $t = 0$ in Eq. (20.179), we find that $F'(+0) = 0$. In addition, we assume that our unknown $F(t)$ has a transform. Transforming Eq. (20.179), using Eqs. (20.147) and (20.148) for the derivatives and Eq. (20.173) to append factors of t , we have

$$-\frac{d}{ds}\left[s^2 f(s) - s\right] + s f(s) - 1 - \frac{d}{ds} f(s) = 0. \quad (20.180)$$

Rearranging and simplifying, we obtain

$$(s^2 + 1)f'(s) + s f(s) = 0,$$

or

$$\frac{df}{f} = -\frac{s ds}{s^2 + 1},$$

a first-order ODE. By integration,

$$\ln f(s) = -\frac{1}{2} \ln(s^2 + 1) + \ln C,$$

which may be rewritten as

$$f(s) = \frac{C}{\sqrt{s^2 + 1}}. \quad (20.181)$$

To confirm that our transform yields the power-series expansion of J_0 , we expand $f(s)$ as given in Eq. (20.181) in a series of negative powers of s , convergent for $s > 1$:

$$\begin{aligned} f(s) &= \frac{C}{s} \left(1 + \frac{1}{s^2}\right)^{-1/2} \\ &= \frac{C}{s} \left[1 - \frac{1}{2s^2} + \frac{1 \cdot 3}{2^2 \cdot 2!s^4} - \cdots + \frac{(-1)^n (2n)!}{(2^n n!)^2 s^{2n}} + \cdots\right]. \end{aligned}$$

Inverting, term by term, we obtain

$$F(t) = C \sum_{n=0}^{\infty} \frac{(-1)^n t^{2n}}{(2^n n!)^2}.$$

When C is set equal to 1, as required by the initial condition $F(0) = 1$, we recover $J_0(t)$, our familiar Bessel function of order zero. Hence,

$$\mathcal{L}\{J_0(t)\} = \frac{1}{\sqrt{s^2 + 1}}. \quad (20.182)$$

This simple, closed form is the Laplace transform of $J_0(t)$. After making a scale change to form $J_0(at)$ using Eq. (20.156), we confirm entry 14 of Table 20.1.

Note that in our derivation of Eq. (20.182) we assumed $s > 1$. The proof for $s > 0$ is the topic of Exercise 20.8.10. ■

It is worth noting that this application was successful and relatively easy because we took $n = 0$ in Bessel's equation. This made it possible to divide out a factor of x (or t). If this had not been done, the terms of the form $t^2 F(t)$ would have introduced a second derivative of $f(s)$. The resulting equation would have been no easier to solve than the original one. This observation illustrates the point that when we go beyond linear ODEs with constant coefficients, the Laplace transform may still be applied, but there is no guarantee that it will be helpful.

The application to Bessel's equation, $n \neq 0$, will be found in the Additional Readings. Alternatively, given the result

$$\mathcal{L}\{J_n(at)\} = \frac{a^{-n}(\sqrt{s^2 + a^2} - s)^n}{\sqrt{s^2 + a^2}}, \quad (20.183)$$

we can confirm its validity by expressing $J_n(t)$ as an infinite series and transforming term by term.

Integration of Transforms

Again, with $F(t)$ at least piecewise continuous and x large enough so that $e^{-xt}F(t)$ decreases exponentially (as $x \rightarrow \infty$), the integral

$$f(x) = \int_0^{\infty} e^{-xt} F(t) dt$$

is uniformly convergent with respect to x . This justifies reversing the order of integration in the following equation:

$$\begin{aligned} \int_s^{\infty} f(x) dx &= \int_s^{\infty} dx \int_0^{\infty} dt e^{-xt} F(t) = \int_0^{\infty} e^{-st} \frac{F(t)}{t} dt, \\ &= \mathcal{L}\left\{\frac{F(t)}{t}\right\}, \end{aligned} \quad (20.184)$$

where the last member of the first line is obtained by integrating with respect to x . The lower limit s must be chosen large enough so that $f(s)$ is within the region of uniform convergence. Equation (20.184) is valid when $F(t)/t$ is finite at $t = 0$ or diverges less strongly than t^{-1} (so that $\mathcal{L}\{F(t)/t\}$ will exist).

For convenience we summarize the definition and properties of the Laplace transform in Table 20.2. Included in the table are formulas for convolution and inversion that will be discussed in Sections 20.9 and Sections 20.10.

Table 20.2 Laplace Transform Operations

	Operation	Equation
1. Laplace transform	$f(s) = \mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) dt$	(15.99)
2. Transform of derivative	$sf(s) - F(+0) = \mathcal{L}\{F'(t)\}$	(15.123)
	$s^2 f(s) - sF(+0) - F'(+0) = \mathcal{L}\{F''(t)\}$	(15.124)
3. Transform of integral	$\frac{1}{s} f(s) = \mathcal{L}\left\{\int_0^t F(x) dx\right\}$	Exercise 20.9.1
4. Change of scale	$\frac{1}{a} f\left(\frac{s}{a}\right) = \mathcal{L}\{F(at)\}$	(20.156)
5. Substitution	$f(s-a) = \mathcal{L}\{e^{at} F(t)\}$	(15.152)
6. Translation	$e^{-bs} f(s) = \mathcal{L}\{F(t-b)\}$	(15.164)
7. Derivative of transform	$f^{(n)}(s) = \mathcal{L}\{(-t)^n F(t)\}$	(15.173)
8. Integral of transform	$\int_s^{\infty} f(x) dx = \mathcal{L}\left\{\frac{F(t)}{t}\right\}$	(15.189)
9. Convolution	$f_1(s)f_2(s) = \mathcal{L}\left\{\int_0^t F_1(t-z)F_2(z) dz\right\}$	(15.193)
10. Inverse transform, Bromwich integral ^a	$\frac{1}{2\pi i} \int_{\beta-i\infty}^{\beta+i\infty} e^{st} f(s) ds = F(t)$	(15.212)

^a β must be large enough that $e^{-\beta t} F(t)$ vanishes as $t \rightarrow +\infty$.

Exercises

20.8.1 Use the expression for the transform of a second derivative to obtain the transform of $\cos kt$.

20.8.2 A mass m is attached to one end of an unstretched spring, spring constant k (Fig. 20.17). Starting at time $t = 0$, the free end of the spring experiences a constant acceleration a , away from the mass. Using Laplace transforms,

- find the position x of m as a function of time.
- determine the limiting form of $x(t)$ for small t .

$$\text{ANS. (a) } x = \frac{1}{2}at^2 - \frac{a}{\omega^2}(1 - \cos \omega t), \quad \omega^2 = \frac{k}{m},$$

$$\text{(b) } x = \frac{a\omega^2}{4!}t^4, \quad \omega t \ll 1.$$

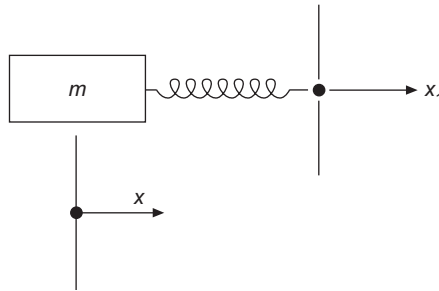


FIGURE 20.17 Spring, Exercise 20.8.2.

20.8.3 Radioactive nuclei decay according to the law

$$\frac{dN}{dt} = -\lambda N,$$

with N the concentration of a given nuclide and λ its particular decay constant. This equation may be interpreted as stating that the rate of decay is proportional to the number of these radioactive nuclei present. They all decay independently.

Consider now a radioactive series of n different nuclides, with Nuclide 1 decaying into Nuclide 2, Nuclide 2 into Nuclide 3, etc., until reaching Nuclide n , which is stable. The concentrations of the various nuclides satisfy the system of ODEs

$$\frac{dN_1}{dt} = -\lambda_1 N_1, \quad \frac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2, \quad \dots, \quad \frac{dN_n}{dt} = \lambda_{n-1} N_{n-1}.$$

- For the case $n = 3$ find $N_1(t)$, $N_2(t)$, and $N_3(t)$, with $N_1(0) = N_0$ and $N_2(0) = N_3(0) = 0$.
- Find an approximate expression for N_2 and N_3 , valid for small t when $\lambda_1 \approx \lambda_2$.
- Find approximate expressions for N_2 and N_3 , valid for large t , when (1) $\lambda_1 \gg \lambda_2$, (2) $\lambda_1 \ll \lambda_2$.

$$\text{ANS. (a) } N_1(t) = N_0 e^{-\lambda_1 t}, \quad N_2(t) = N_0 \frac{\lambda_1}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}),$$

$$N_3(t) = N_0 \left(1 - \frac{\lambda_2}{\lambda_2 - \lambda_1} e^{-\lambda_1 t} + \frac{\lambda_1}{\lambda_2 - \lambda_1} e^{-\lambda_2 t} \right).$$

$$\text{(b) } N_2 \approx N_0 \lambda_1 t, \quad N_3 \approx \frac{N_0}{2} \lambda_1 \lambda_2 t^2.$$

$$\text{(c) (1) } N_2 \approx N_0 e^{\lambda_2 t}$$

$$N_3 \approx N_0 (1 - e^{-\lambda_2 t}), \quad \lambda_1 t \gg 1.$$

$$\text{(2) } N_2 \approx N_0 (\lambda_1 / \lambda_2) e^{-\lambda_1 t},$$

$$N_3 \approx N_0 (1 - e^{-\lambda_1 t}), \quad \lambda_2 t \gg 1.$$

20.8.4 The rate of formation of an isotope in a nuclear reactor is given by

$$\frac{dN_2}{dt} = \varphi \left[\sigma_1 N_1(0) - \sigma_2 N_2(t) \right] - \lambda_2 N_2(t).$$

Here $N_1(0)$ is the concentration of the original isotope (assumed constant), and N_2 is that of the newly formed isotope. The first two terms on the right-hand side describe the production and destruction of the new isotope via neutron absorption; φ is the neutron flux (units $\text{cm}^{-2}\text{s}^{-1}$); σ_1 and σ_2 (units cm^2) are neutron absorption cross sections. The final term describes the radioactive decay of the new isotope, with decay constant λ_2 .

- Find the concentration N_2 of the new isotope as a function of time.
- For original isotope ^{153}Eu , $\sigma_1 = 400$ barns $= 400 \times 10^{-24} \text{ cm}^2$, $\sigma_2 = 1000$ barns $= 1000 \times 10^{-24} \text{ cm}^2$, and $\lambda_2 = 1.4 \times 10^{-9} \text{ s}^{-1}$. If $N_1(0) = 10^{20}$ and $\varphi = 10^9 \text{ cm}^{-2}\text{s}^{-1}$, find N_2 , the concentration of ^{154}Eu , after 1 year of continuous irradiation. Is the assumption that N_1 is constant justified?

20.8.5 In a nuclear reactor ^{135}Xe is formed as both a direct fission product of ^{235}U and by decay of ^{135}I (another fission product), half-life 6.7 hours. The half-life of ^{135}Xe is 9.2 hours. Because ^{135}Xe strongly absorbs thermal neutrons, thereby “poisoning” the nuclear reactor, its concentration is a matter of great interest. The relevant equations are

$$\begin{aligned} \frac{dN_I}{dt} &= \varphi \gamma_I (\sigma_f N_U) - \lambda_I N_I, \\ \frac{dN_{\text{Xe}}}{dt} &= \varphi \left[\gamma_{\text{Xe}} (\sigma_f N_U) - \sigma_{\text{Xe}} N_{\text{Xe}} \right] + \lambda_I N_I - \lambda_{\text{Xe}} N_{\text{Xe}}. \end{aligned}$$

Here N_I , N_{Xe} , N_U are the concentrations of ^{135}I , ^{135}Xe , ^{235}U , with N_U assumed to be constant. The neutron flux φ in the reactor causes fission of ^{235}U with cross section σ_f and removes ^{135}Xe by neutron absorption with cross section $\sigma_{\text{Xe}} = 3.5 \times 10^6$ barns $= 3.5 \times 10^{-18} \text{ cm}^2$. Neutron absorption by ^{135}I is negligible. The yield of ^{135}I and ^{135}Xe per fission are, respectively, $\gamma_I = 0.060$ and $\gamma_{\text{Xe}} = 0.003$.

- Find $N_{\text{Xe}}(t)$ in terms of neutron flux φ and the product $\sigma_f N_U$.
- Find $N_{\text{Xe}}(t \rightarrow \infty)$.
- After N_{Xe} has reached equilibrium, the reactor is shut down: $\varphi = 0$. Find $N_{\text{Xe}}(t)$ following shutdown. Note the short-term increase in N_{Xe} , which may for a few hours interfere with starting the reactor up again.

Hint. The half-life $t_{1/2}$ of a radioactive isotope is the time required for decay of half of the nuclides in a sample. For a decay rate $dN/dt = -\lambda N$, the half-life has the value $t_{1/2} = \ln 2 / \lambda$, so λ can be computed as $\lambda = \ln 2 / t_{1/2} = 0.693 / t_{1/2}$.

20.8.6 Solve Eq. (20.160), which describes a damped simple harmonic oscillator, for $X(0) = X_0$, $X'(0) = 0$, and

- $b^2 = 4mk$ (critically damped),
- $b^2 > 4mk$ (overdamped).

ANS. (a) $X(t) = X_0 e^{-(b/2m)t} \left(1 + \frac{b}{2m} t \right).$

20.8.7 Again solve Eq. (20.160), which describes a damped simple harmonic oscillator, but this time for $X(0) = 0$, $X'(0) = v_0$, and

- (a) $b^2 < 4mk$ (underdamped),
- (b) $b^2 = 4mk$ (critically damped),
- (c) $b^2 > 4mk$ (overdamped).

ANS. (a) $X(t) = \frac{v_0}{\omega_1} e^{-(b/2m)t} \sin \omega_1 t$,
 (b) $X(t) = v_0 t e^{-(b/2m)t}$.

20.8.8 The motion of a body falling in a resisting medium may be described by

$$m \frac{d^2 X(t)}{dt^2} = mg - b \frac{dX(t)}{dt}$$

when the retarding force is proportional to the velocity. Find $X(t)$ and $dX(t)/dt$ for the initial conditions

$$X(0) = \frac{dX}{dt} \Big|_{t=0} = 0.$$

20.8.9 Ringing circuit. In certain electronic devices, resistance, inductance, and capacitance are placed in a circuit as shown in Fig. 20.18. A constant voltage is maintained across the capacitance, keeping it charged. At time $t = 0$ the circuit is disconnected from the voltage source. Find the voltages across each of the elements R , L , and C as a function of time. Assume R to be small.

Hint. By Kirchhoff's laws

$$I_{RL} + I_C = 0 \quad \text{and} \quad E_R + E_L = E_C,$$

where

$$E_R = I_{RL} R, \quad E_L = L \frac{dI_{RL}}{dt}, \quad E_C = \frac{q_0}{C} + \frac{1}{C} \int_0^t I_C dt,$$

q_0 = initial charge of capacitor.

20.8.10 With $J_0(t)$ expressed as a contour integral, apply the Laplace transform operation, reverse the order of integration, and thus show that

$$\mathcal{L}\{J_0(t)\} = (s^2 + 1)^{-1/2}, \quad \text{for } s > 0.$$

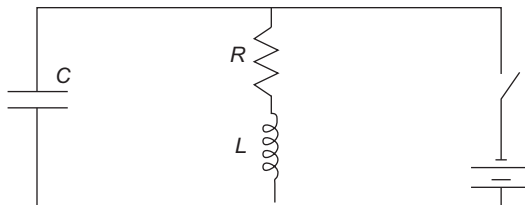


FIGURE 20.18 Ringing circuit.

- 20.8.11** Develop the Laplace transform of $J_n(t)$ from $\mathcal{L}\{J_0(t)\}$ by using the Bessel function recurrence relations.

Hint. Here is a chance to use mathematical induction (Section 1.4).

- 20.8.12** A calculation of the magnetic field of a circular current loop in circular cylindrical coordinates leads to the integral

$$\int_0^{\infty} e^{-kz} k J_1(ka) dk, \quad \Re z \geq 0.$$

Show that this integral is equal to $a/(z^2 + a^2)^{3/2}$.

- 20.8.13** Show that

$$\mathcal{L}\{I_0(at)\} = (s^2 - a^2)^{-1/2}, \quad s > a.$$

- 20.8.14** Verify the following Laplace transforms:

- (a) $\mathcal{L}\{j_0(at)\} = \mathcal{L}\left\{\frac{\sin at}{at}\right\} = \frac{1}{a} \cot^{-1}\left(\frac{s}{a}\right),$
 (b) $\mathcal{L}\{y_0(at)\}$ does not exist.
 (c) $\mathcal{L}\{i_0(at)\} = \mathcal{L}\left\{\frac{\sinh at}{at}\right\} = \frac{1}{2a} \ln \frac{s+a}{s-a} = \frac{1}{a} \coth^{-1}\left(\frac{s}{a}\right),$
 (d) $\mathcal{L}\{k_0(at)\}$ does not exist.

- 20.8.15** Develop a Laplace transform solution of Laguerre's equation,

$$tF''(t) + (1-t)F'(t) + nF(t) = 0.$$

Note that you need a derivative of a transform and a transform of derivatives. Go as far as you can with a general value of n ; then (and only then) set $n = 0$.

- 20.8.16** Show that the Laplace transform of the Laguerre polynomial $L_n(at)$ is given by

$$\mathcal{L}\{L_n(at)\} = \frac{(s-a)^n}{s^{n+1}}, \quad s > 0.$$

- 20.8.17** Show that

$$\mathcal{L}\{E_1(t)\} = \frac{1}{s} \ln(s+1), \quad s > 0,$$

where

$$E_1(t) = \int_t^{\infty} \frac{e^{-\tau}}{\tau} d\tau = \int_1^{\infty} \frac{e^{-xt}}{x} dx.$$

$E_1(t)$ is the exponential integral function, first encountered in this book in Table 1.2.

20.8.18 (a) From Eq. (20.184) show that

$$\int_0^{\infty} f(x) dx = \int_0^{\infty} \frac{F(t)}{t} dt,$$

provided the integrals exist.

(b) From the preceding result show that

$$\int_0^{\infty} \frac{\sin t}{t} dt = \frac{\pi}{2},$$

in agreement with Eqs. (20.146) and (11.107).

20.8.19 (a) Show that

$$\mathcal{L} \left\{ \frac{\sin kt}{t} \right\} = \cot^{-1} \left(\frac{s}{k} \right).$$

(b) Using this result (with $k = 1$), prove that

$$\mathcal{L} \{ \text{si}(t) \} = -\frac{1}{s} \tan^{-1} s,$$

where

$$\text{si}(t) = -\int_t^{\infty} \frac{\sin x}{x} dx, \quad \text{the sine integral.}$$

20.8.20 If $F(t)$ is periodic (Fig. 20.19) with a period a so that $F(t + a) = F(t)$ for all $t \geq 0$, show that

$$\mathcal{L} \{ F(t) \} = \frac{1}{1 - e^{-as}} \int_0^a e^{-st} F(t) dt.$$

Note that the integration is now over only the **first period** of $F(t)$.

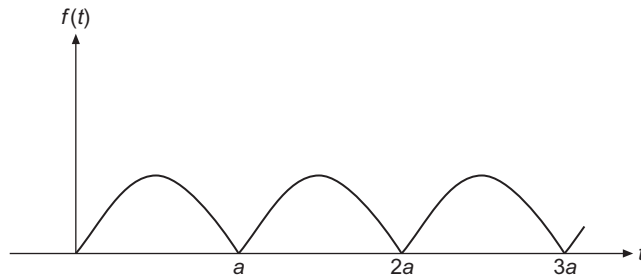


FIGURE 20.19 Periodic function.

20.8.21 Find the Laplace transform of the square wave (period a) defined by

$$F(t) = \begin{cases} 1, & 0 < t < a/2, \\ 0, & a/2 < t < a. \end{cases}$$

$$\text{ANS. } f(s) = \frac{1}{s} \frac{1 - e^{-as/2}}{1 - e^{-as}}.$$

20.8.22 Show that

$$(a) \quad \mathcal{L}\{\cosh at \cos at\} = \frac{s^3}{s^4 + 4a^4},$$

$$(b) \quad \mathcal{L}\{\cosh at \sin at\} = \frac{as^2 + 2a^3}{s^4 + 4a^4},$$

$$(c) \quad \mathcal{L}\{\sinh at \cos at\} = \frac{as^2 - 2a^3}{s^4 + 4a^4},$$

$$(d) \quad \mathcal{L}\{\sinh at \sin at\} = \frac{2a^2s}{s^4 + 4a^4}.$$

20.8.23 Show that

$$(a) \quad \mathcal{L}^{-1}\{(s^2 + a^2)^{-2}\} = \frac{1}{2a^3} \sin at - \frac{t}{2a^2} \cos at,$$

$$(b) \quad \mathcal{L}^{-1}\{s(s^2 + a^2)^{-2}\} = \frac{t}{2a} \sin at,$$

$$(c) \quad \mathcal{L}^{-1}\{s^2(s^2 + a^2)^{-2}\} = \frac{1}{2a} \sin at + \frac{t}{2} \cos at,$$

$$(d) \quad \mathcal{L}^{-1}\{s^3(s^2 + a^2)^{-2}\} = \cos at - \frac{at}{2} \sin at.$$

20.8.24 Show that

$$\mathcal{L}\{(t^2 - k^2)^{-1/2} u(t - k)\} = K_0(ks).$$

Hint. Try transforming an integral representation of $K_0(ks)$ into the Laplace transform integral.

20.9 LAPLACE CONVOLUTION THEOREM

One of the most important properties of the Laplace transform is that given by the convolution, or Faltung, theorem. We take two transforms,

$$f_1(s) = \mathcal{L}\{F_1(t)\} \quad \text{and} \quad f_2(s) = \mathcal{L}\{F_2(t)\},$$

and multiply them together:

$$f_1(s)f_2(s) = \int_0^\infty e^{-sx} F_1(x) dx \int_0^\infty e^{-sy} F_2(y) dy. \quad (20.185)$$

If we introduce the new variable $t = x + y$ and integrate over t and y instead of x and y , the limits of integration become $(0 \leq t \leq \infty)$, $(0 \leq y \leq t)$. Noting that the Jacobian of the transformation from (x, y) to (t, y) is unity, we have

$$\begin{aligned} f_1(s)f_2(s) &= \int_0^\infty e^{-st} dt \int_0^t F_1(t-y)F_2(y)dy \\ &= \mathcal{L} \left\{ \int_0^t F_1(t-y)F_2(y)dy \right\} \\ &= \mathcal{L}\{F_1 * F_2\}, \end{aligned} \quad (20.186)$$

where, similarly to the Fourier transform, we use the notation

$$\int_0^t F_1(t-z)F_2(z)dz \equiv F_1 * F_2, \quad (20.187)$$

and call this operation the **convolution** of F_1 and F_2 . It can be shown that convolution is symmetric:

$$F_1 * F_2 = F_2 * F_1. \quad (20.188)$$

Carrying out the inverse transform, we also find

$$\mathcal{L}^{-1}\{f_1(s)f_2(s)\} = \int_0^t F_1(t-z)F_2(z)dz = F_1 * F_2. \quad (20.189)$$

Convolution formulas are useful for finding new transforms or, in some cases, as an alternative to a partial fraction expansion. They also find use in the solution of integral equations, as is illustrated in Chapter 21.

Example 20.9.1 DRIVEN OSCILLATOR WITH DAMPING

As one illustration of the use of the convolution theorem, let us return to the mass m on a spring, with damping and a driving force $F(t)$. The equation of motion, Eq. (20.160), now becomes

$$mX''(t) + bX'(t) + kX(t) = F(t). \quad (20.190)$$

Initial conditions $X(0) = 0$, $X'(0) = 0$ are used to simplify this illustration, and the transformed equation is

$$ms^2x(s) + bsx(s) + kx(s) = f(s),$$

with solution

$$x(s) = \frac{f(s)}{m} \frac{1}{(s + b/2m)^2 + \omega_1^2}, \quad (20.191)$$

where, as in [Example 20.8.5](#),

$$\omega_0^2 \equiv \frac{k}{m}, \quad \omega_1^2 \equiv \omega_0^2 - \frac{b^2}{4m^2}. \quad (20.192)$$

We identify the right-hand side of [Eq. \(20.191\)](#) as the product of two known transforms:

$$\frac{f(s)}{m} = \frac{1}{m} \mathcal{L}\{F(t)\}, \quad \frac{1}{(s + b/2m)^2 + \omega_1^2} = \frac{1}{\omega_1} \mathcal{L}\{e^{-(b/2m)t} \sin \omega_1 t\},$$

where the second of these is a case of [Eq. \(20.158\)](#).

Now applying the convolution theorem, [Eq. \(20.189\)](#), we obtain the solution to our original problem as an integral:

$$X(t) = \mathcal{L}^{-1}\{x(s)\} = \frac{1}{m\omega_1} \int_0^t F(t-z) e^{-(b/2m)z} \sin \omega_1 z \, dz. \quad (20.193)$$

We go on to consider two specific choices for the driving force $F(t)$. We first take the impulsive force $F(t) = P\delta(t)$. Then

$$X(t) = \frac{P}{m\omega_1} e^{-(b/2m)t} \sin \omega_1 t. \quad (20.194)$$

Here P represents the momentum transferred by the impulse, and the constant P/m takes the place of an initial velocity $X'(0)$.

As a second case, let $F(t) = F_0 \sin \omega t$. We could again use [Eq. \(20.193\)](#), but a partial fraction expansion is perhaps more convenient. With

$$f(s) = \frac{F_0 \omega}{s^2 + \omega^2}$$

[Eq. \(20.191\)](#) can be written in the partial fraction form,

$$\begin{aligned} x(s) &= \frac{F_0 \omega}{m} \frac{1}{s^2 + \omega^2} \frac{1}{(s + b/2m)^2 + \omega_1^2} \\ &= \frac{F_0 \omega}{m} \left[\frac{a's + b'}{s^2 + \omega^2} + \frac{c's + d'}{(s + b/2m)^2 + \omega_1^2} \right], \end{aligned} \quad (20.195)$$

with coefficients a' , b' , c' , and d' (independent of s) to be determined. Direct calculation shows for a' and b'

$$\begin{aligned} -\frac{1}{a'} &= \frac{b}{m} \omega^2 + \frac{m}{b} (\omega_0^2 - \omega^2)^2, \\ -\frac{1}{b'} &= -\frac{m}{b} (\omega_0^2 - \omega^2) \left[\frac{b}{m} \omega^2 + \frac{m}{b} (\omega_0^2 - \omega^2)^2 \right]. \end{aligned}$$

The terms of $x(s)$ containing a' and b' lead upon inversion of the Laplace transform to the steady-state component of the solution:

$$X(t) = \frac{F_0}{[b^2 \omega^2 + m^2 (\omega_0^2 - \omega^2)^2]^{1/2}} \sin(\omega t - \varphi), \quad (20.196)$$

where

$$\tan \varphi = \frac{b\omega}{m(\omega_0^2 - \omega^2)}.$$

Differentiating the denominator, we find that the amplitude has a maximum when $\omega = \omega_2$, with

$$\omega_2^2 = \omega_0^2 - \frac{b^2}{2m^2} = \omega_1^2 - \frac{b^2}{4m^2}. \quad (20.197)$$

This is the resonance condition.¹² At resonance the amplitude becomes $F_0/b\omega_1$, showing that the mass m goes into infinite oscillation at resonance if damping is neglected ($b = 0$).

This calculation differs from those used for the determination of transfer functions (compare [Example 20.5.1](#)) in that a steady-state solution at a fixed frequency is not assumed. Use of the Laplace transform (rather than the Fourier transform) permits solution for transient as well as steady-state components of the solution. The transients, which we will not work out in detail, arise from the terms of [Eq. \(20.195\)](#) involving c' and d' . These terms contain the quantity $(s + b/2m)^2$ in the denominator, and its presence will generate terms of the inverse transform that contain the exponential factor $e^{-bt/2m}$. In other words, these terms describe exponentially decaying transients.

It is worth noting that we have had three different characteristic frequencies:

$$\text{Resonance for forced oscillations with damping: } \omega_2^2 = \omega_0^2 - \frac{b^2}{2m^2},$$

$$\text{Free oscillation frequency, with damping: } \omega_1^2 = \omega_0^2 - \frac{b^2}{4m^2},$$

$$\text{Free oscillation frequency, no damping: } \omega_0^2 = \frac{k}{m}.$$

These frequencies coincide only if the damping is zero. ■

Recall that [Eq. \(20.190\)](#) is our ODE for the response of a dynamical system to an arbitrary driving force. The final response clearly depends on both the driving force and the characteristics of our system. This dual dependence is separated in the transform space. In [Eq. \(20.191\)](#) the transform of the response (output) appears as the product of two factors, one describing the driving force (input) and the other describing the dynamical system. This is a factorization similar to that we found when discussing the use of Fourier transforms in signal-processing applications in [Section 20.5](#).

Exercises

20.9.1 From the convolution theorem show that

$$\frac{1}{s} f(s) = \mathcal{L} \left\{ \int_0^t F(x) dx \right\},$$

where $f(s) = \mathcal{L}\{F(t)\}$.

¹²The amplitude (squared) has the typical resonance denominator (the Lorentz line shape), found in [Exercise 20.2.8](#).

20.9.2 If $F(t) = t^a$ and $G(t) = t^b$, $a > -1$, $b > -1$,

(a) Show that the convolution $F * G$ is given by

$$F * G = t^{a+b+1} \int_0^1 y^a (1-y)^b dy.$$

(b) By using the convolution theorem, show that

$$\int_0^1 y^a (1-y)^b dy = \frac{a! b!}{(a+b+1)!} = B(a+1, b+1),$$

where B is the beta function.

20.9.3 Using the convolution integral, calculate

$$\mathcal{L}^{-1} \left\{ \frac{s}{(s^2 + a^2)(s^2 + b^2)} \right\}, \quad a^2 \neq b^2.$$

20.9.4 An undamped oscillator is driven by a force $F_0 \sin \omega t$. Find the displacement $X(t)$ as a function of time, subject to initial conditions $X(0) = X'(0) = 0$. Note that the solution is a linear combination of two simple harmonic motions, one with the frequency of the driving force and one with the frequency ω_0 of the free oscillator.

$$\text{ANS. } X(t) = \frac{F_0/m}{\omega^2 - \omega_0^2} \left(\frac{\omega}{\omega_0} \sin \omega_0 t - \sin \omega t \right).$$

20.10 INVERSE LAPLACE TRANSFORM

Bromwich Integral

We now develop an expression for the inverse Laplace transform $\mathcal{L}^{-1}$ appearing in the equation

$$F(t) = \mathcal{L}^{-1} \{ f(s) \}. \quad (20.198)$$

One approach lies in the Fourier transform, for which we know the inverse relation. There is a difficulty, however. Our Fourier transformable function had to satisfy the Dirichlet conditions. In particular, we required that in order for $g(\omega)$ to be a valid Fourier transform,

$$\lim_{\omega \rightarrow \infty} g(\omega) = 0, \quad (20.199)$$

so that the infinite integral would be well defined.¹³ Now we wish to treat functions $F(t)$ that may diverge exponentially. To surmount this difficulty, we extract an exponential factor, $e^{\beta t}$, from our (possibly) divergent $F(t)$ and write

$$F(t) = e^{\beta t} G(t). \quad (20.200)$$

¹³We made an exception to deal with the delta function, but even in that case $g(\omega)$ was bounded for all ω .

If $F(t)$ diverges as $e^{\alpha t}$, we require β to be greater than α so that $G(t)$ will be **convergent**. Now, with $G(t) = 0$ for $t < 0$ and otherwise suitably restricted so that it may be represented by a Fourier integral, as in Eq. (20.22), we have

$$G(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iut} du \int_0^{\infty} G(v) e^{-iuv} dv. \quad (20.201)$$

Inserting Eq. (20.201) into Eq. (20.200), we have

$$F(t) = \frac{e^{\beta t}}{2\pi} \int_{-\infty}^{\infty} e^{iut} du \int_0^{\infty} F(v) e^{-\beta v} e^{-iuv} dv. \quad (20.202)$$

We now make a change of variable to $s = \beta + iu$, causing the integral over v in Eq. (20.202) to assume the form of a Laplace transform:

$$\int_0^{\infty} F(v) e^{-sv} dv = f(s).$$

The variable s is now complex, but must be restricted to $\Re(s) \geq \beta$ in order to guarantee convergence. Note that the Laplace transform has extended a function specified on the positive real axis onto the complex plane, $\Re s \geq \beta$.¹⁴

We now need to rewrite Eq. (20.202) using the variable s in place of u . The range $-\infty < u < \infty$ corresponds to a contour in the complex plane of s , which is a vertical line from $\beta - i\infty$ to $\beta + i\infty$; we also need to substitute $du = ds/i$. Making these changes, Eq. (20.202) becomes

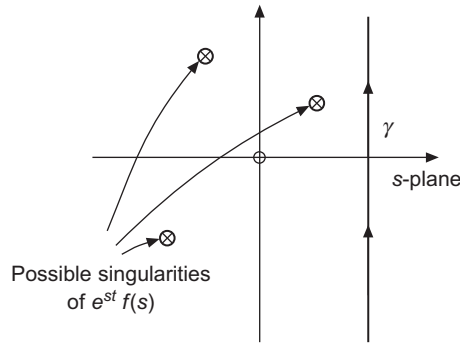
$$F(t) = \frac{1}{2\pi i} \int_{\beta - i\infty}^{\beta + i\infty} e^{st} f(s) ds. \quad (20.203)$$

Here is our **inverse transform**. The path has become an infinite vertical line in the complex plane. Note that the constant β was chosen so that $f(s)$ would be nonsingular for $s \geq \beta$. It can be shown that the nonsingularity of $f(s)$ extends to complex s provided that $\Re s \geq \beta$, so the integrand of Eq. (20.203) can have singularities only to the left of the integration path. See Fig. 20.20.

The inverse transformation given by Eq. (20.203) is known as the **Bromwich integral**, although sometimes it is referred to as the **Fourier-Mellin theorem** or **Fourier-Mellin integral**. This integral may now be evaluated by the regular methods of contour integration (Chapter 11). If $t > 0$ and $f(s)$ is analytic except for isolated singularities (and no branch points), and is also small at large $|s|$, the contour may be closed by an infinite semicircle in the left half-plane that does not contribute to the integral. Then by the residue theorem (Section 11.8),

$$F(t) = \sum (\text{residues included for } \Re s < \beta). \quad (20.204)$$

¹⁴For a derivation of the inverse Laplace transform using only real variables, see C. L. Bohn and R. W. Flynn, Real variable inversion of Laplace transforms: An application in plasma physics, *Am. J. Phys.* **46**: 1250 (1978).

FIGURE 20.20 Possible singularities of $e^{st}f(s)$.

It is worth mentioning that in many cases of interest $f(s)$ may become large in the left half-plane or have branch points, and evaluation of the Bromwich integral may then present significant challenges.

Possibly this means of evaluation with $\Re s$ ranging through negative values seems paradoxical in view of our previous requirement that $\Re s \geq \beta$. The paradox disappears when we recall that the requirement $\Re s \geq \beta$ was imposed to guarantee convergence of the Laplace transform integral that defined $f(s)$. Once $f(s)$ is obtained, we may then proceed to exploit its properties as an analytic function in the complex plane wherever we choose.

Perhaps a pair of examples may clarify the evaluation of Eq. (20.203).

Example 20.10.1 INVERSION VIA CALCULUS OF RESIDUES

If $f(s) = a/(s^2 - a^2)$, then the integrand for the Bromwich integral will be

$$e^{st}f(s) = \frac{ae^{st}}{s^2 - a^2} = \frac{ae^{st}}{(s + a)(s - a)}. \quad (20.205)$$

From the form of Eq. (20.205), we see that this integrand has poles at $s = \pm a$, and the value of β for the integral must be larger than $|a|$. Since these are simple poles, it is easy to verify that the residue at $s = a$ must be $e^{at}/2$, while the residue at $s = -a$ will be $-e^{-at}/2$. The form of the integrand also permits us to close the contour in the left half-plane. We find, in accord with Eq. (20.204),

$$\text{Residues} = \left(\frac{1}{2}\right)(e^{at} - e^{-at}) = \sinh at = F(t). \quad (20.206)$$

Equation (20.206) is in agreement with entry #7 of our table of Laplace transforms, Table 20.1. ■

Example 20.10.2 MULTIREGION INVERSION

If $f(s) = (1 - e^{-as})/s$, the Bromwich integral then has integrand

$$e^{st}f(s) = e^{st} \left(\frac{1 - e^{-as}}{s} \right), \quad (20.207)$$

and the possibilities for closing the contour depend on the relative magnitudes of t and a .

Considering first $t > a$, we may close the contour for the Bromwich integral in the left half-plane without changing its value. Our integrand is an entire function (analytic everywhere in the finite s -plane; note that the s in the denominator cancels when the numerator is expanded in a Maclaurin series). Since no singularities are enclosed, we conclude that for $t > a$, $F(t) = 0$.

For t in the range $0 < t < a$, a different situation is encountered. Expanding the integrand into the two terms

$$\frac{e^{st}}{s} - \frac{e^{s(t-a)}}{s},$$

we see that the first becomes small in the left half-plane (but large in the right half-plane), while the second term behaves in an opposite fashion (large in the left half-plane, small in the right). The obvious solution is to use different contours for the two terms, each of which is individually singular, with a pole at $s = 0$. We therefore close the contour for the first term in the left half-plane, but close that for the second term in the right half-plane. Since the vertical portion of the contour is at $\Re s = \beta > 0$, we see that the integral of the first term encloses the singularity, while the integral of the second term does not. Therefore the first integral will have a value equal to the residue of the integrand at the singularity (this residue is 1), while the second integral will vanish. These contours are illustrated in Fig. 20.21.

Finally, for $t < 0$, the entire integrand becomes small in the right half-plane, the contour (for the entire integrand) surrounds no singularities, and the integral is zero. Summarizing these three cases,

$$F(t) = u(t) - u(t - a) = \begin{cases} 0, & t < 0, \\ 1, & 0 < t < a, \\ 0, & t > a, \end{cases} \quad (20.208)$$

a step function of unit height and length a (Fig. 20.22). ■

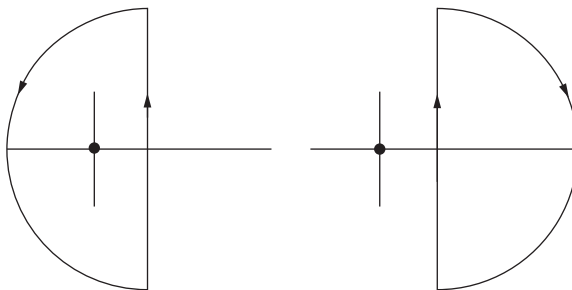
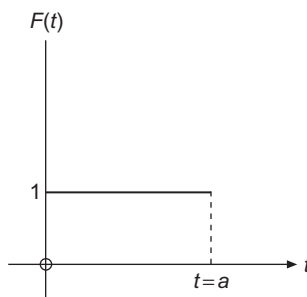


FIGURE 20.21 Contours for Example 20.10.2.

FIGURE 20.22 Finite-length step function $u(t) - u(t - a)$.

Two general comments may be in order. First, these two examples hardly begin to show the usefulness and power of the Bromwich integral. It is always available for inverting a complicated transform when the tables prove inadequate.

Second, this derivation is not presented as a rigorous one. Rather, it is given more as a plausibility argument, although it can be made rigorous. The determination of the inverse transform is somewhat similar to the solution of a differential equation. It makes little difference how you get the inverse transform. Guess at it if you want. It can always be checked by verifying that

$$\mathcal{L}\{F(t)\} = f(s).$$

Two alternate derivations of the Bromwich integral are the subjects of [Exercises 20.10.1](#) and [\(20.10.2\)](#).

Exercises

20.10.1 Derive the Bromwich integral from Cauchy's integral formula.

Hint. Apply the inverse transform $\mathcal{L}^{-1}$ to

$$f(s) = \frac{1}{2\pi i} \lim_{\alpha \rightarrow \infty} \int_{\beta - i\alpha}^{\beta + i\alpha} \frac{f(z)}{s - z} dz,$$

where $f(z)$ is analytic for $\Re z \geq \beta$.

20.10.2 Starting with

$$\frac{1}{2\pi i} \int_{\beta - i\infty}^{\beta + i\infty} e^{st} f(s) ds,$$

show that by introducing

$$f(s) = \int_0^{\infty} e^{-sz} F(z) dz$$

we can convert our integral into the Fourier representation of a Dirac delta function. From this derive the inverse Laplace transform.

20.10.3 Derive the Laplace transformation convolution theorem by use of the Bromwich integral.

20.10.4 Find

$$\mathcal{L}^{-1} \left\{ \frac{s}{s^2 - k^2} \right\}$$

- (a) by a partial fraction expansion.
- (b) Repeat, using the Bromwich integral.

20.10.5 Find

$$\mathcal{L}^{-1} \left\{ \frac{k^2}{s(s^2 + k^2)} \right\}$$

- (a) by using a partial fraction expansion.
- (b) Repeat using the convolution theorem.
- (c) Repeat using the Bromwich integral.

ANS. $F(t) = 1 - \cos kt$.

20.10.6 Use the Bromwich integral to find the function whose transform is $f(s) = s^{-1/2}$. Note that $f(s)$ has a branch point at $s = 0$. The negative x -axis may be taken as a cut line. See Fig. 20.23.

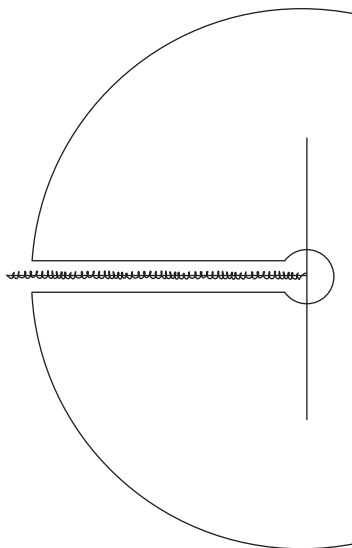


FIGURE 20.23 Contour for Exercise 20.10.6.

Hint. A portion of the path needed to close the contour will yield nonzero contributions to the contour integral. These will need to be taken into account to get the proper value for the Bromwich integral.

$$\text{ANS. } F(t) = (\pi t)^{-1/2}.$$

20.10.7 Show that

$$\mathcal{L}^{-1} \left\{ (s^2 + 1)^{-1/2} \right\} = J_0(t)$$

by evaluation of the Bromwich integral.

Hint. Convert your Bromwich integral into an integral representation of $J_0(t)$. [Figure 20.24](#) shows a possible contour.

20.10.8 Evaluate the inverse Laplace transform

$$\mathcal{L}^{-1} \left\{ (s^2 - a^2)^{-1/2} \right\}$$

by each of the following methods:

- (a) Expansion in a series and term-by-term inversion.
- (b) Direct evaluation of the Bromwich integral.
- (c) Change of variable in the Bromwich integral: $s = (a/2)(z + z^{-1})$.

20.10.9 Show that

$$\mathcal{L}^{-1} \left\{ \frac{\ln s}{s} \right\} = -\ln t - \gamma,$$

where $\gamma = 0.5772 \dots$ is the Euler-Mascheroni constant.

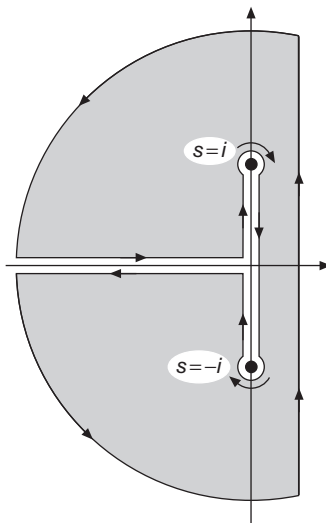


FIGURE 20.24 A possible contour for the inversion of $J_0(t)$.

20.10.10 Evaluate the Bromwich integral for

$$f(s) = \frac{s}{(s^2 + a^2)^2}.$$

20.10.11 Heaviside expansion theorem. If the transform $f(s)$ may be written as a ratio

$$f(s) = \frac{g(s)}{h(s)},$$

where $g(s)$ and $h(s)$ are analytic functions, with $h(s)$ having simple, isolated zeros at $s = s_i$, show that

$$F(t) = \mathcal{L}^{-1} \left\{ \frac{g(s)}{h(s)} \right\} = \sum_i \frac{g(s_i)}{h'(s_i)} e^{s_i t}.$$

Hint. See Exercise 11.6.3.

20.10.12 Using the Bromwich integral, invert $f(s) = s^{-2}e^{-ks}$. Express $F(t) = \mathcal{L}^{-1}\{f(s)\}$ in terms of the (shifted) unit step function $u(t - k)$.

$$ANS. \quad F(t) = (t - k)u(t - k).$$

20.10.13 You have a Laplace transform:

$$f(s) = \frac{1}{(s + a)(s + b)}, \quad a \neq b.$$

Invert this transform by each of three methods:

- (a) Partial fractions and use of tables,
- (b) Convolution theorem,
- (c) Bromwich integral.

$$ANS. \quad F(t) = \frac{e^{-bt} - e^{-at}}{a - b}, \quad a \neq b.$$

Additional Readings

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